The Sum - Installation Instructions

Note: *throughout these instructions,* ***your AWS region might get changed*** *when you go from page to page. If you cannot find your application or environment in the AWS console, it is probably because your region (which you can set in the top right corner of any page next to your user) is not set to the same region as where you made your App. If you change this, then your app should appear.*

Note: *the installation instructions do not cover resetting the database for migration issues, rebuilding the environment and redeploying, ssh, and creating a super user. For this information, see the user manual.*

Note: *you will be charged by AWS for these resources. If you wish to no longer wish to be charged,* ***you must delete the resources and take down your application****. There are instructions at the end of this document for how to delete the S3 bucket, database, environment, and application. If you are a student testing these instructions, Professor Ibrahim said he should be able to reimburse you for any of these costs.*

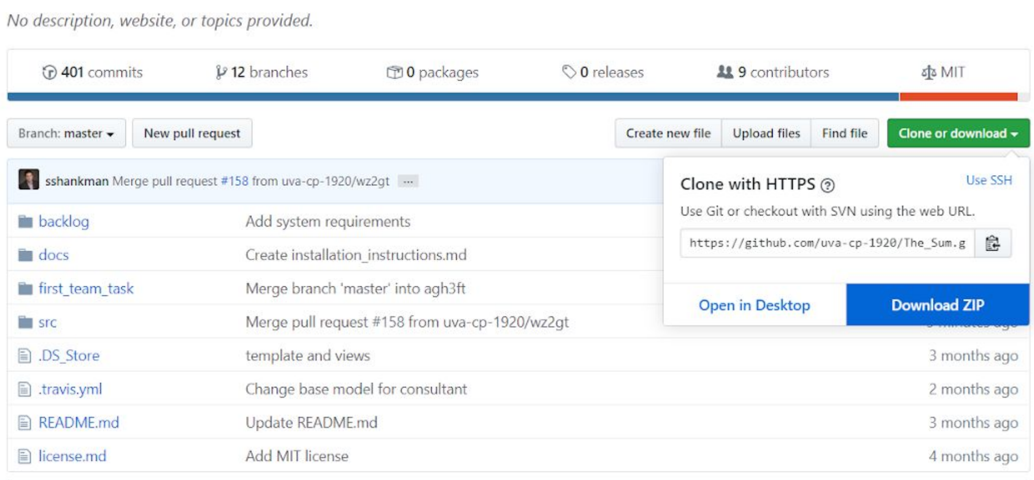
Before you begin, copy paste the following table somewhere convenient. You will be filling it out throughout the process, and will need it during deployment:

Name	Value
ENV	PRODUCTION
EMAIL_HOST	email-smtp.us-east-1.amazonaws.com
ERROR_REPORT_EMAIL	Any email you want
APP_URL	**This comes from part 17a**
RDS_DB_NAME	**This comes from part 18f**
PRODUCTION_DB_PASSWORD	**This comes from part 13c**
RDS_USER	**This comes from part 13b**
RDS_HOST	**This comes from part 18g**
RDS_PORT	**This comes from part 18h**

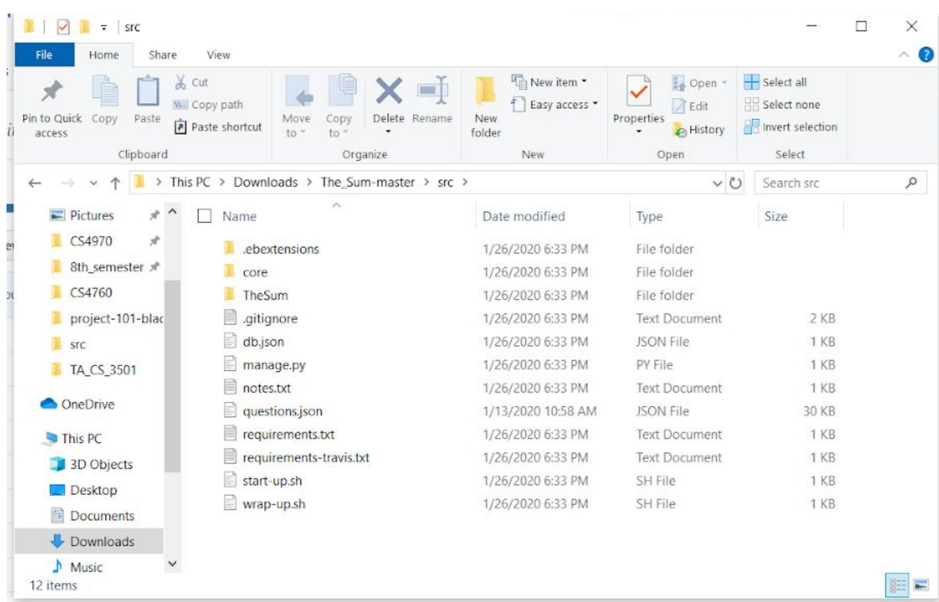
SECRET_KEY	uC9OvT@rJ9&sAj&3G980cAp0g$hEe5izR?H59D^VRhCr7r3uTS *Note: you may also pick some arbitrary long piece of text here*
AWS_ACCESS_KEY_ID	**This comes from part 7g or you should already know it**
AWS_SECRET_ACCESS_KEY	**This comes from part 7g or you should already know it**
AWS_STORAGE_BUCKET_NAME	**This comes from part 22a**
AWS_S3_REGION_NAME	**This comes from part 22c**
PAYPAL_IDENTITY_TOKEN	**This comes from part 25**
PAYPAL_RECIEVER_EMAIL	**This comes from part 23**
SEND_EMAIL_PDF	**This comes from part 30**
FROM_EMAIL	info@thepowerofdifference.org
EMAIL_BACKEND_PROD	django_ses.SESBackend
BCC_EMAIL	powerofdifferencetest@gmail.com (test) *Note: this should be whatever email address you want to receive the BCC for the PDF results*

Initial Deployment

1. For the most up to date code, if you have access to it, you should download the code from our GitHub repo! **If you do not have access to the GitHub repo, go to step 2**
    a. On the master branch page, click clone or download and then click download zip:

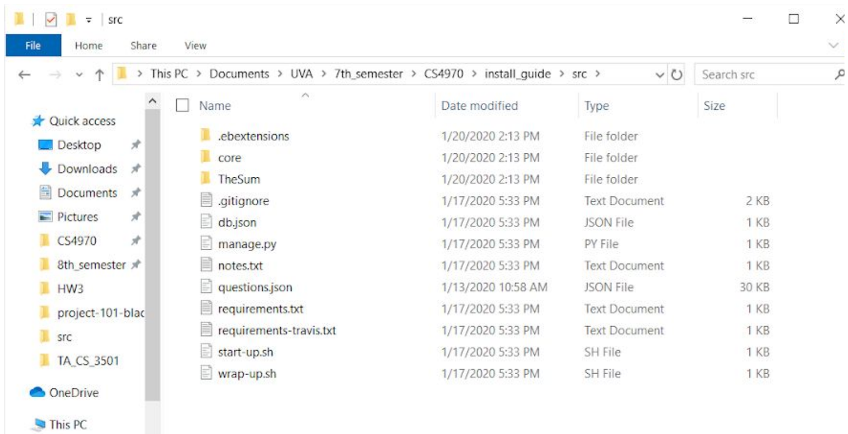


b. Unzip this source code and enter your new unzipped folder. From there, you must enter the folder called “src” (if you only see one folder and it is not called “src”, enter that and repeat until you find the “src” folder)
c. Download the questions.json file here: https://drive.google.com/file/d/1rPeHbhr0XBgJQ5pU-W2fuTTfCJH2XAK_/view?usp=sharing
    i. You must be using a valid UVA email for this to work
    ii. If you are our customer (The Sum), then we have already given you the questions.json file. It is not in the GitHub repo as you asked us to keep it private. Please use the questions.json file we have previously given you instead of downloading it.
d. Put the questions.json file into the “src” folder
e. You should see the following:

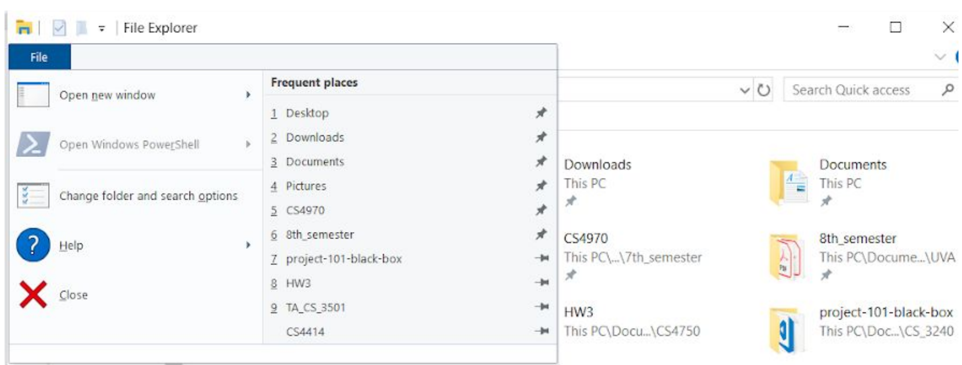


f. Go to step 3

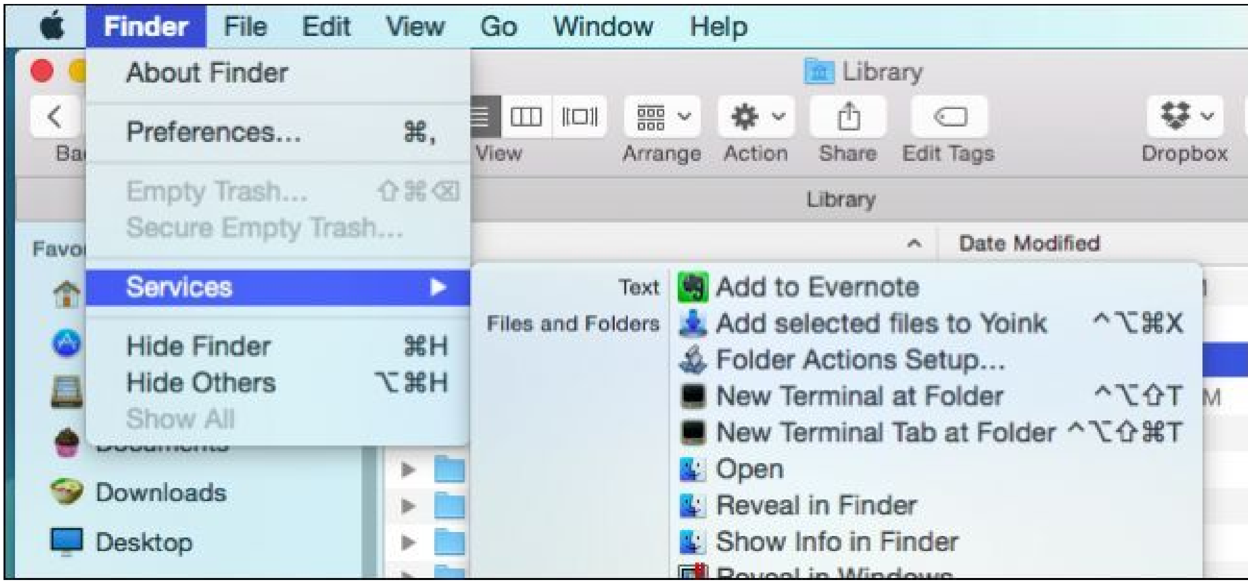
2. Download the source code from https://drive.google.com/open?id=1ARIBuAHOPS42jwHMzaKHdoOdsq2uthOY
    a. You must be using a valid UVA email for this to work
    b. Unzip this source code and go into your new unzipped folder
        i. **Note: you should unzip this into a new folder or else the installation may fail**
    c. You should see the following:



3. Open a terminal (command line) in this folder (the one that you navigated to in step 1 or step 2)
    a. In the terminal, after typing each command, you must press enter
    b. You can often open a terminal in a folder like in the following images:
    c. Windows:

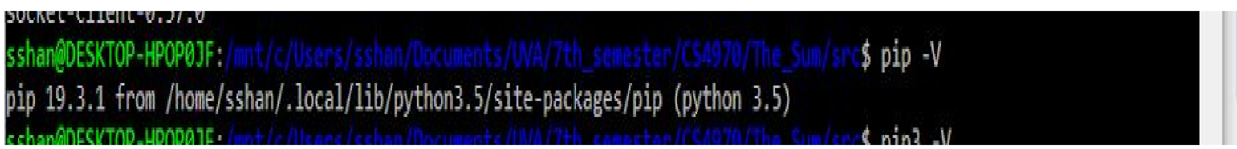


    d. MAC - there is a New Terminal at Folder option:

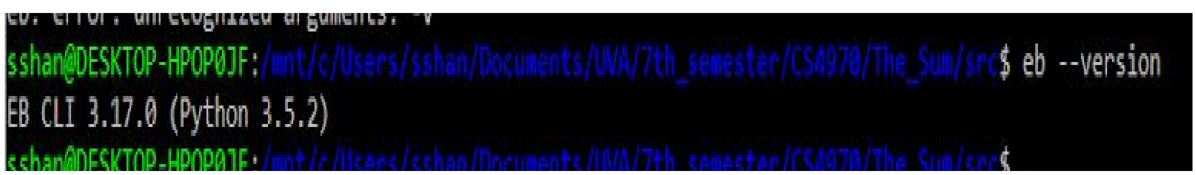


e. If you want to get a terminal into this folder in any other method, then that is fine. Just make sure you are within the folder and the files you see are like those from step 2a

4. If you do not have git installed, follow this guide here: https://git-scm.com/book/en/v2/Getting-Started-Installing-Git
5. If you do not have pip installed, follow this guide here: http://pip.pypa.io/en/stable/installing/
    a. You can check if you have pip installed by the following (in this case, pip is installed)

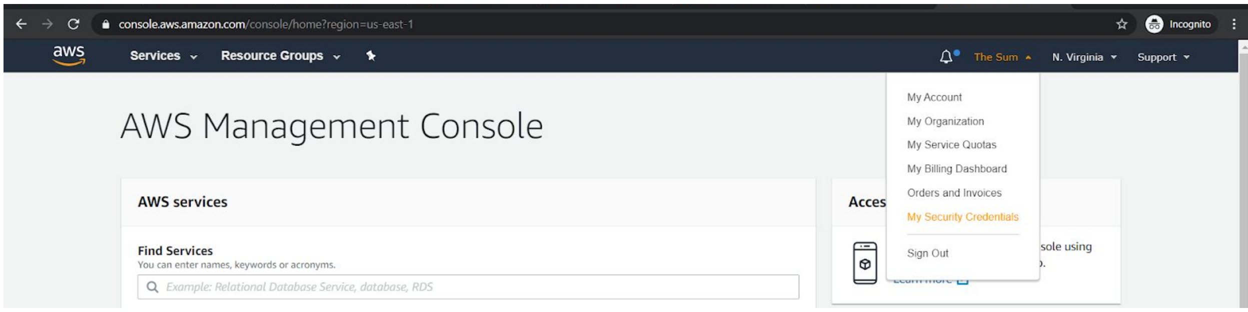


6. If you do not have awsebcli installed, follow the instructions here: https://pypi.org/project/awsebcli/
    a. To check if you have it installed:



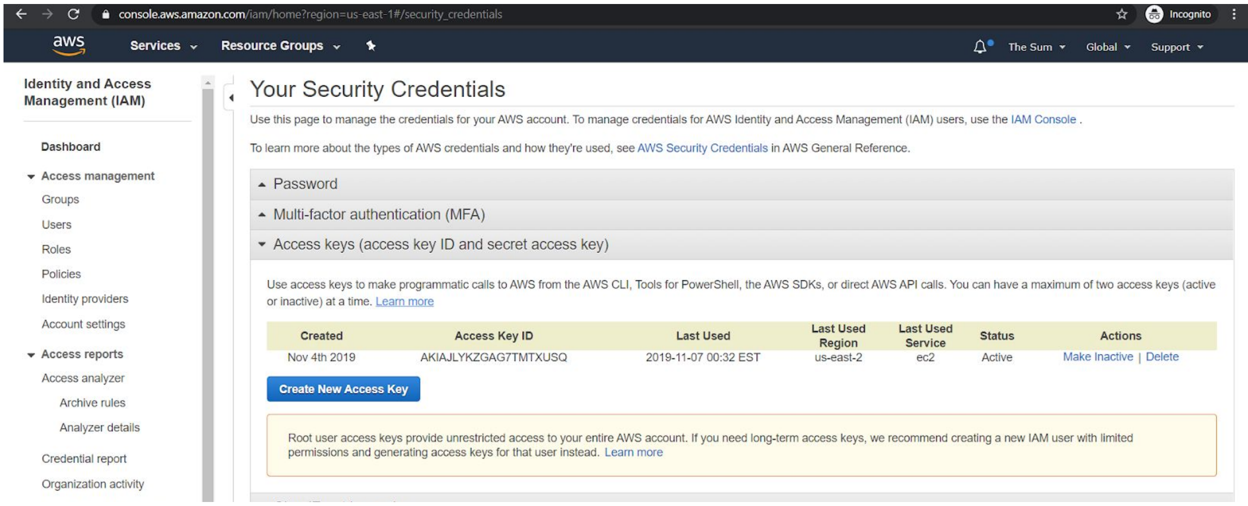
7. Make sure you already have and AWS account and access credentials
    a. If you are not sure, go to https://aws.amazon.com/ and click on “sign in to the console” in the top right corner
        i. If needed, make a new account

b. Click on your account name and then go to “My security credentials” (image below)



c. If asked, say “continue to security credentials”

d. Go to “Access Keys”



e. If you already have an access key here and you know both the access key ID and secret key, then you should be able to use those. **In the table that you copied at the beginning, use these to fill out the AWS_ACCESS_KEY_ID, and AWS_SECRET_ACCESS_KEY fields respectively.** Otherwise follow these next few steps.

f. Click “Create New Access Key”

g. Next, click “Download Key File”

   i. **Keep this file safe. It is the only way you can get your secret key!**

   ii. **Open the file. You should see an Access Key ID and a secret access key. In the table that you copied at the beginning, use the Access Key ID as the value for AWS_ACCESS_KEY_ID, and the Secret Access Key as the value for AWS_SECRET_ACCESS_KEY**

8. If you already have an AWS account registered with the awsebcli (what we installed in step 6), then you will have to edit the file located at ~/.aws/config . You can do this with whatever text editor you want. Typically we use vim. To do so with vim:

   a. Have a terminal open

b. Type “vim ~/.aws/config” and press enter
    i. In vim, you can edit the text by typing “i”. You get out of edit mode, just press escape. To save the file after exiting edit mode, type “:x” and then press enter
c. Edit to add a new profile and number it as the next number that you need.
d. Edit the file to make it look like the following (actual codes have been turned into XXXX for secrutiy purposes in this guide):
Here [profile eb-cli] is your pre-existing profile you’re using for any other projects and the [profile eb-cli2] is the one you will be using for this project.

```
[profile eb-cli]
aws_access_key_id = XXXXXXXXXXXXX
aws_secret_access_key = XXXXXXXXXXXX

[profile eb-cli2]
aws_access_key_id = XXXXXXXXXX
aws_secret_access_key = XXXXXXXXXXXX
```

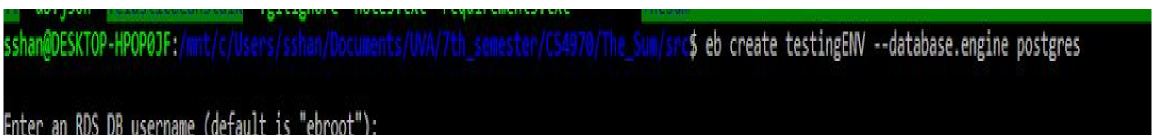
9. If you had to add a new profile in ~/.aws/config, then all the eb commands will require --profile eb-cli2 (or whatever you named the new profile linked to the AWS you want to use) to follow them. For example, in step 10, we will use “`eb init`”. If you added profile eb-cli 2, you will need to type “`eb init --profile eb-cli2`”
10. Go back to the terminal we opened in step 3 and run ‘”`eb init`”’
    a. This will ask you to choose a region. Select a region that you think will be central to the users of your application. We recommend choosing 1 (us-east-1). Type 1 (or the number of whatever region you choose) and press enter
    b. You will then be asked for your AWS credentials (type them in when prompted for each part)
11. You will see something like the following:

```
Select an application to use
1) debugging
2) The Sum
3) [ Create new Application ]
(default is 3):
```

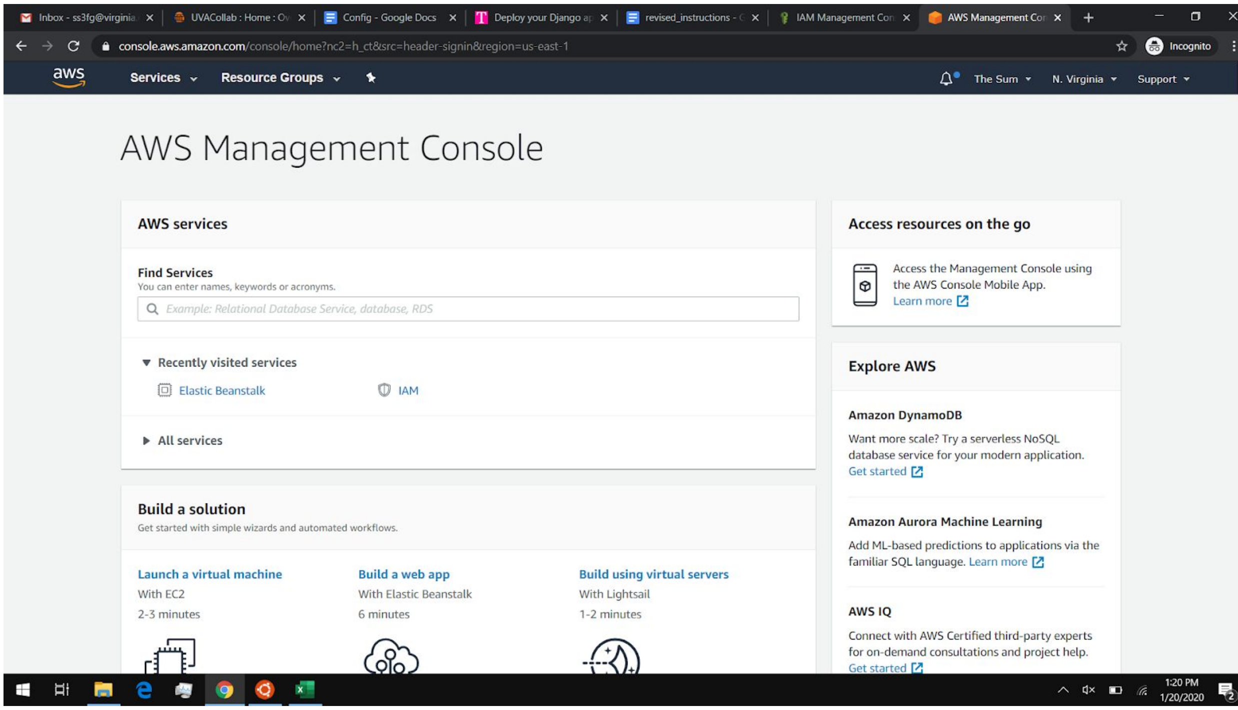
    a. Type the number for [Create new Application] and press enter
    b. You will be asked for a name. Give the application a name a press enter

    c. You should be asked if you are using Python. Type Y and press enter
    d. You will be asked about which Python version. To use the default (which we recommend), type 1 and press enter.
    e. You should be asked to set up SSH. Type Y and press enter.
    f. You will be asked to make a key pair name. Use the default of aws-eb (and just press enter to use this, do not type anything)
        i. If you already have a keypair you want to use or you want to make a new one, then follow the prompts to do so
    g. You will be asked for a key passphrase, choose whatever you want and **remember it**! You may simply just type enter to have no passphrase

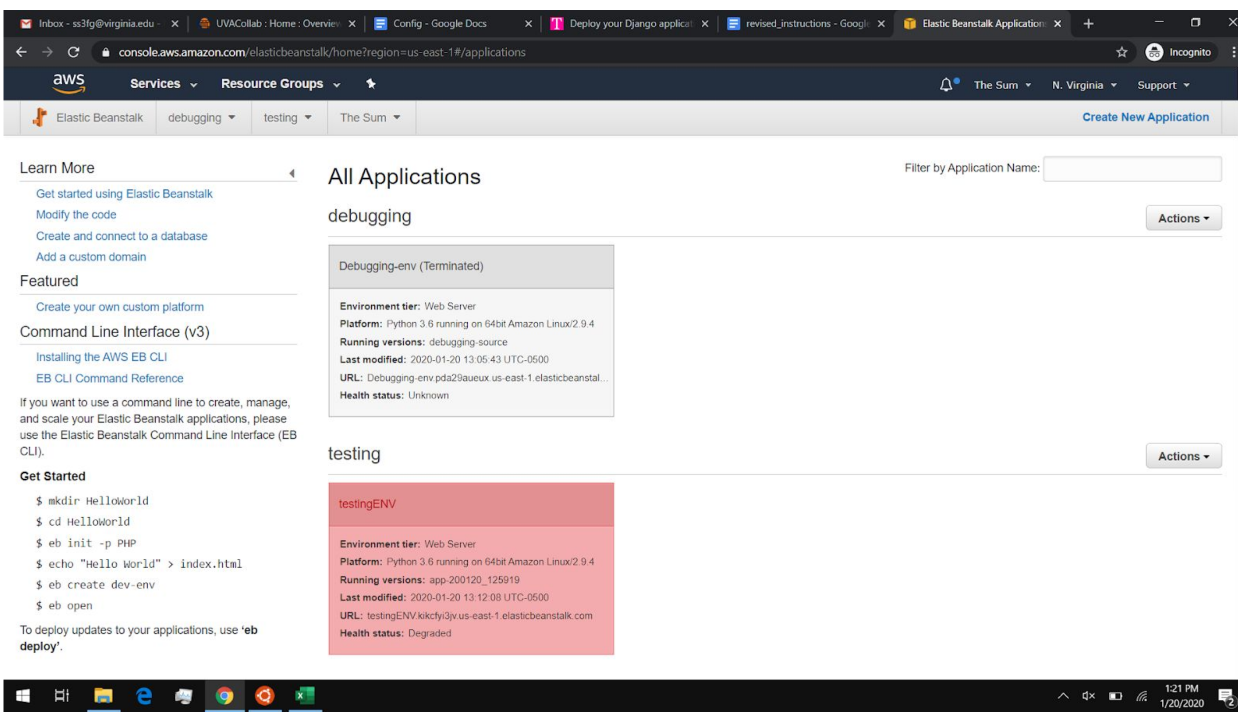
12. Now we want to create our elastic beanstalk (AWS) environment! You will want to use a name that ends in ENV (for readability purposes). In this example, we will use testingENV for our environment
13. Run "`eb create <name from step 12> --database.engine postgres`"
    a. Image:



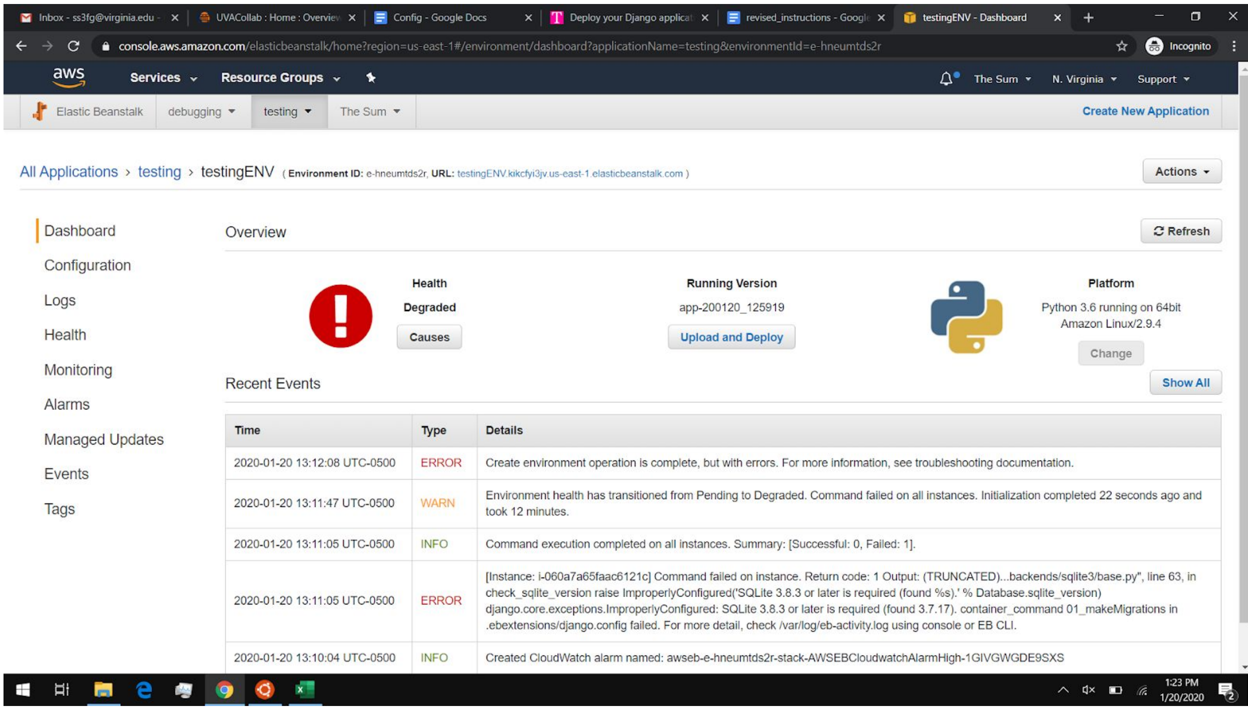
    b. You will be asked for an RDS DB username. Either type a username that you choose and press enter or press enter to use the default which is ebroot . **In the table you copied at the beginning, fill this out as the value for RDS_USER**
    c. You will then be asked for an RDS DB master password. Type one in and press enter. **In the table you copied at the beginning, fill this out as the value for PRODUCTION_DB_PASSWORD**
    d. This will take a while to run as AWS must set up our database. Let it run. This may take 20 or more minutes.
    e. As long as the 'Create Environment Operation is complete' message shows up, you can disregard errors related to 'Environment Variables'. We will fix those later.
14. After this has finished, you should see the "$" prompt again.
15. Open a web browser and go to https://aws.amazon.com/ and login
    a. You should see a page like the following:



16. In find services, type in “elastic beanstalk”
    a. Only one option should appear. Click on it! You should now see a page like the following:



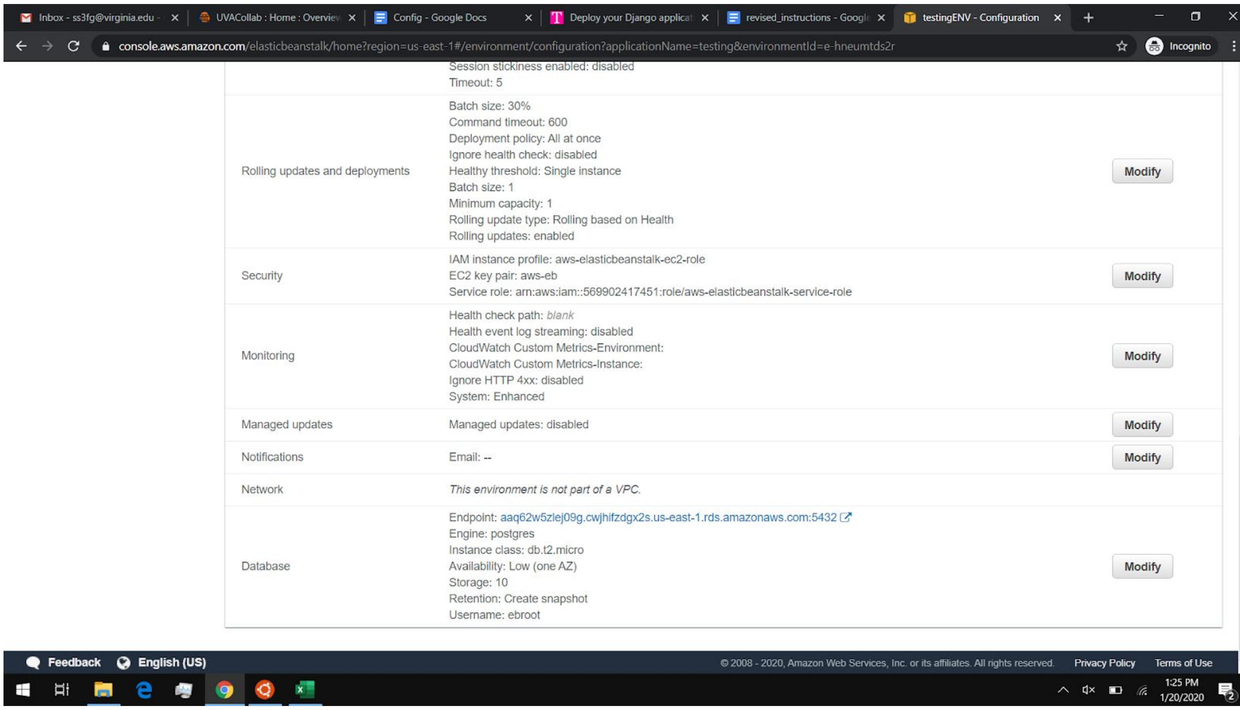
17. In this guide, we called our application testing and our environment testingENV. Click on the part where you see testingENV (in the image above it is in red). You should now see a page like the following:



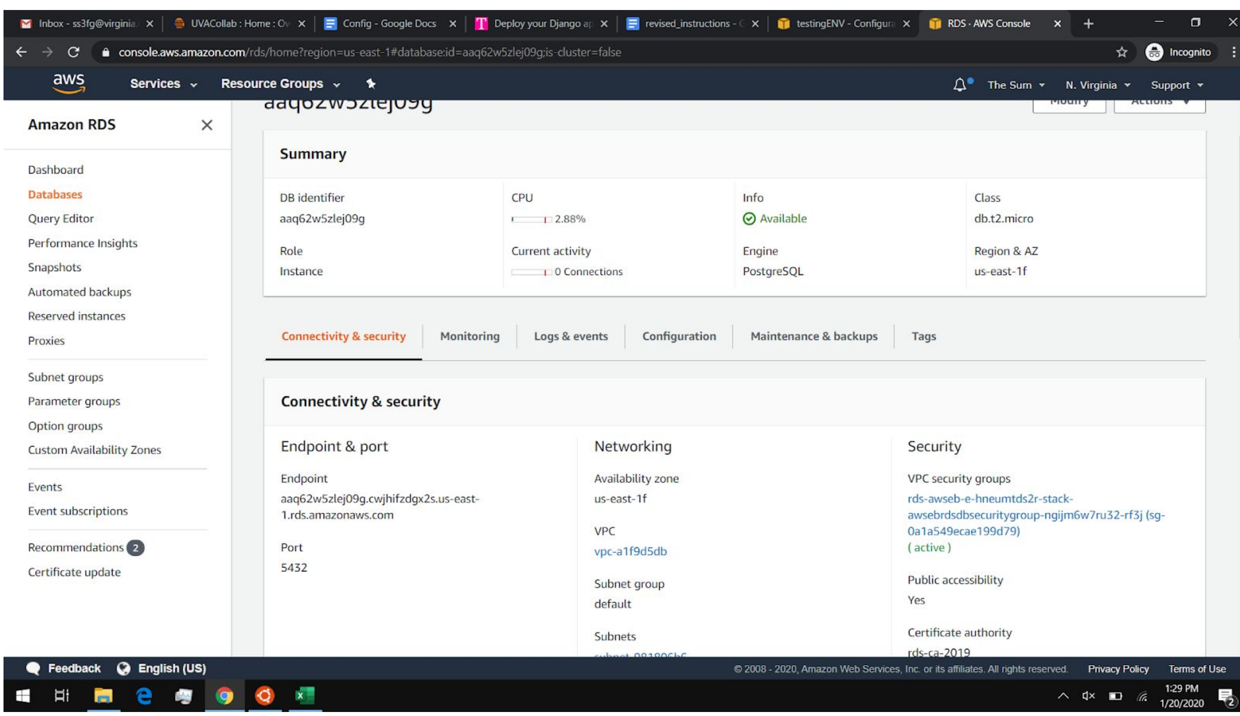
a. On the top bar where you see All Applications > testing > testingENV, you should see a URL. **In the table you copied at the beginning, fill this as the value for APP_URL (do not include http://… only use exactly what the URL on this page says)**
b. In the end, you should be able to find your application at this URL
c. On the left hand sidebar, click configuration

Getting the Database Info

18. Scroll down until you see “Database” (like the following image)



a. Look at the endpoint. Remember everything before the first "." *(In the case of the screenshot, it's aaq62w5zlej09g)*
b. Open the link next to endpoint in a new tab
c. Click on the DB identifier that matches what you remembered in step 18 a
d. From here you should see the following:



e. You will need the info on this page to fill out some of the fields in the table you copied at the beginning.

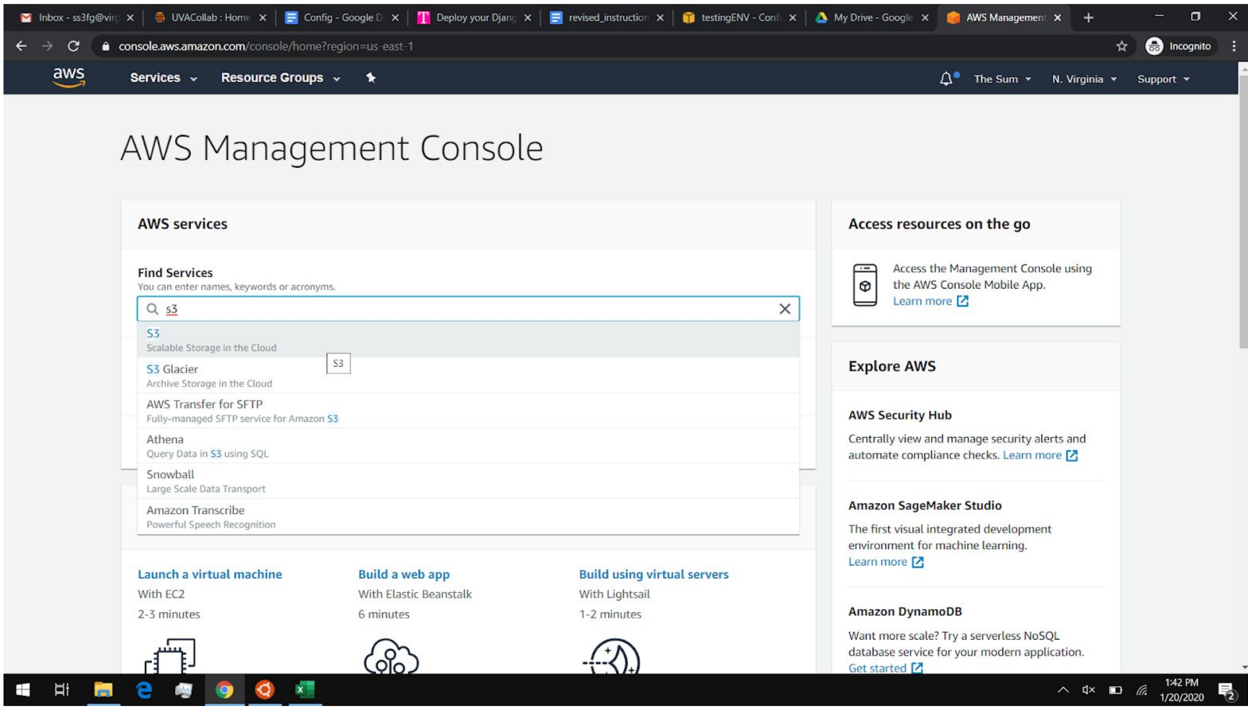
f. **DB identifier:** fill this out as the value for **RDS_DB_NAME.** *(In the case of the screenshot, it's aaq62w5zlej09g)*

g. **Endpoint**: (this is a bit different than the endpoint from the previous pages) fill this out as the value for **RDS_HOST** *(In the case of the screenshot, it's aaq62w5zlej09g.cwjhfzdgx2us.us-east-1.rds.amazonaws.com)*

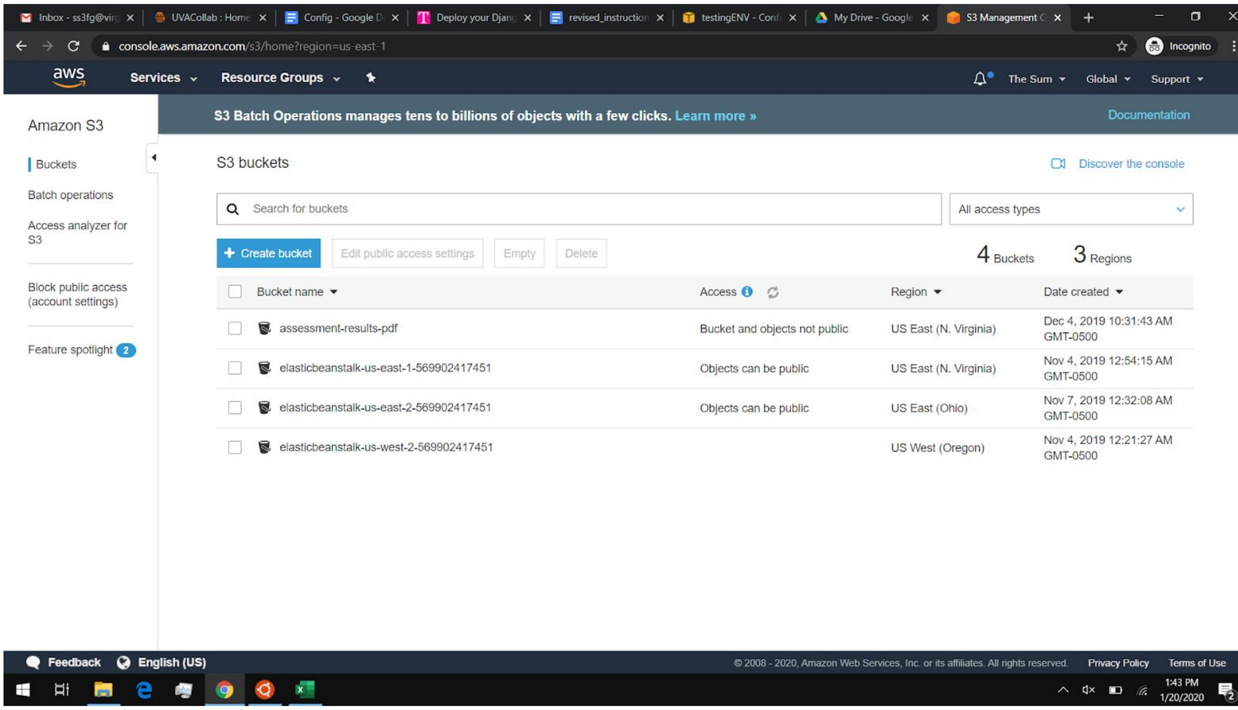
h. **Port**: fill this out as the value for **RDS_PORT** *(In the case of the screenshot, it's 5432)*

Create S3 Bucket for static files

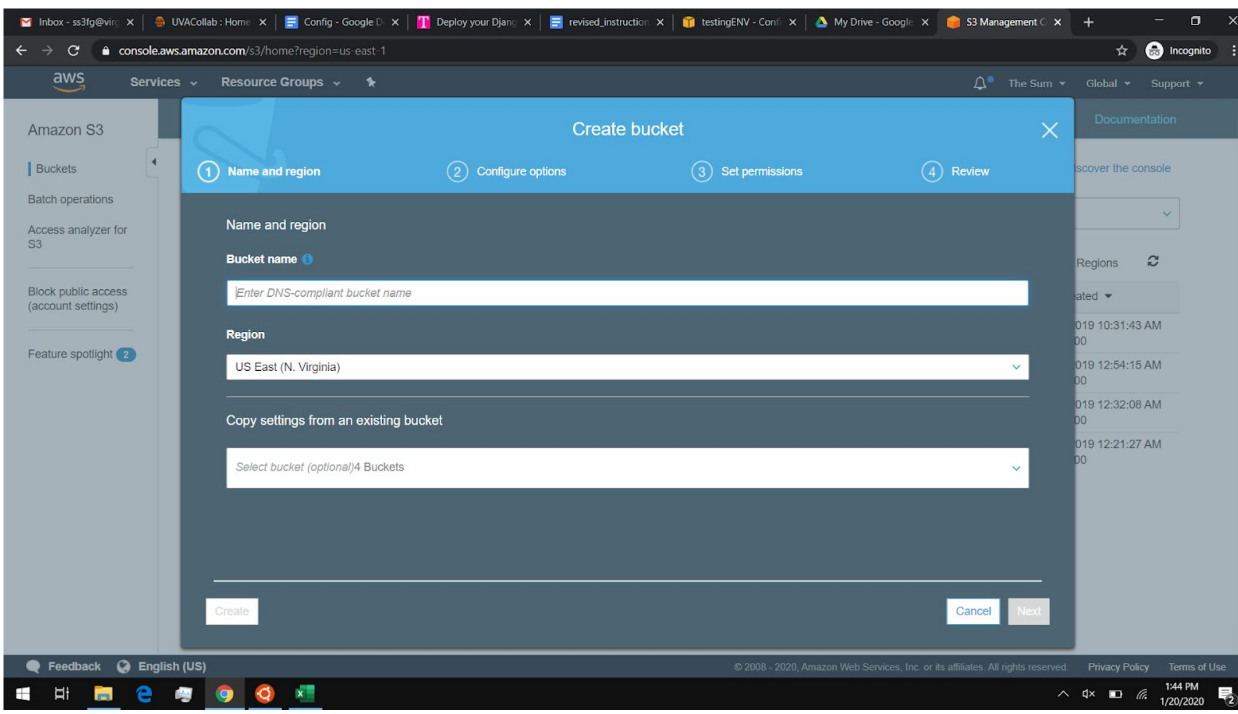
19. Now go to https://aws.amazon.com/ and get to to management console like in step 15. If you know you are already logged in, you can go directly to https://console.aws.amazon.com/
20. In Find Services, type S3 and click on the option that only says S3 (image below)



21. You should see a screen like the following:



22. Click on “Create Bucket”



a. For bucket name, type in a name that gets accepted and that you will use as the name for your S3 bucket. **In the table you copied at the beginning, fill this out as the value for AWS_STORAGE_BUCKET_NAME**

b. For region, we recommend using US East (N. Virginia)

c. Regardless of which region you choose, you need to get the region name as you need it for the **AWS_S3_REGION_NAME** environment variable
    i. For example, If you choose 'US East (N.Virginia), then your region name will be 'us-east-1' and you will then use 'us-east-1' as your environment variable for AWS_S3_REGION_NAME
    ii. Mapping Table:

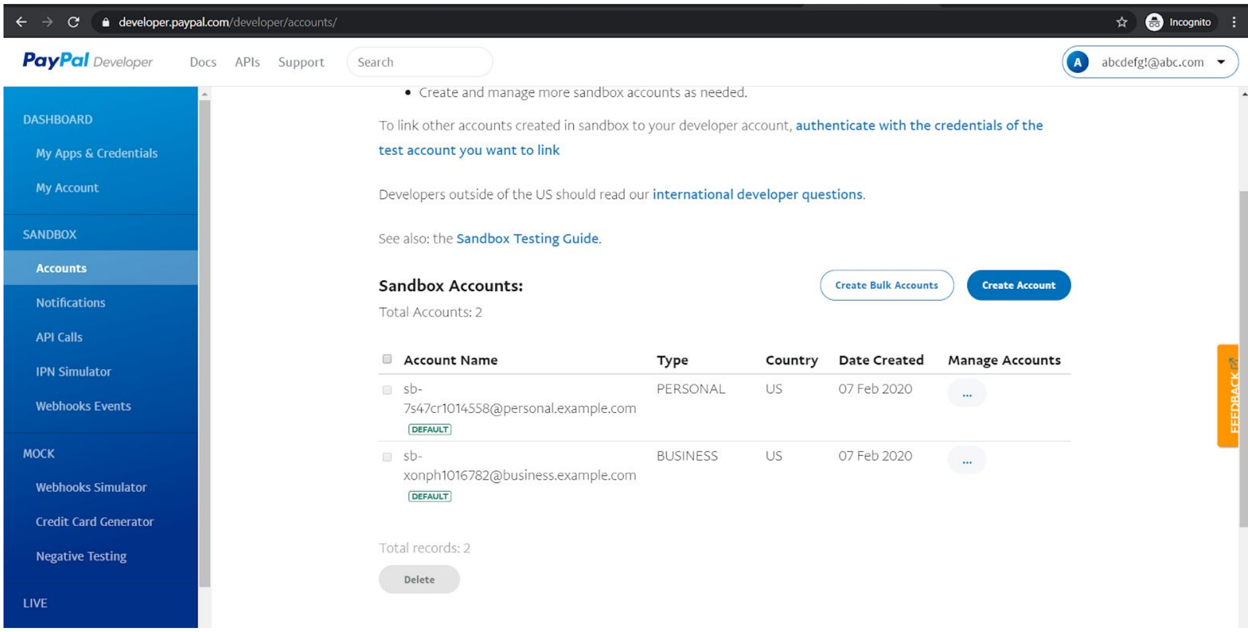
Region	Name
US East (Ohio)	us-east-2
US East (N. Virginia)	us-east-1
US West (N. California)	us-west-1
US West (Oregon)	us-west-2
Asia Pacific (Hong Kong)	ap-east-1
Asia Pacific (Mumbai)	ap-south-1
Asia Pacific (Osaka-Local)	ap-northeast-3
Asia Pacific (Seoul)	ap-northeast-2
Asia Pacific (Singapore)	ap-southeast-1
Asia Pacific (Sydney)	ap-southeast-2
Asia Pacific (Tokyo)	ap-northeast-1
Canada (Central)	ca-central-1

Europe (Frankfurt)	eu-central-1
Europe (Ireland)	eu-west-1
Europe (London)	eu-west-2
Europe (Paris)	eu-west-3
Europe (Stockholm)	eu-north-1
Middle East (Bahrain)	me-south-1
South America (São Paulo)	sa-east-1

   d. Keep clicking next and then eventually click create bucket
   e. You should now see your new bucket listed

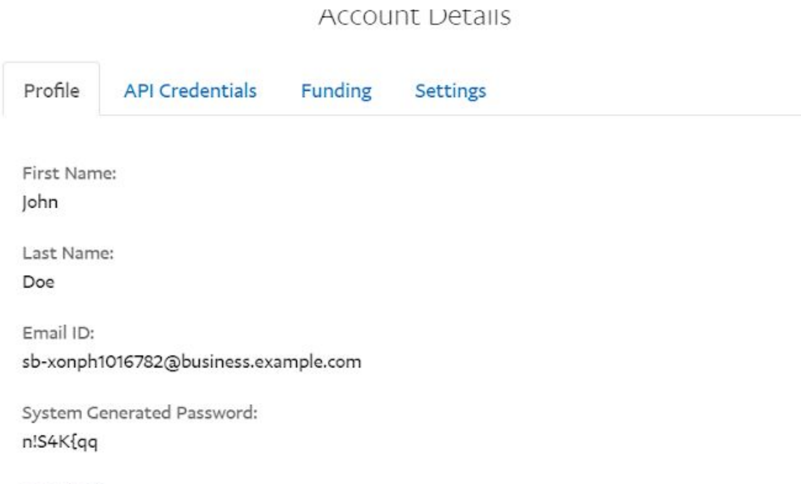
PayPal

23. Sign up (if needed) and log in to your paypal **business account**: https://www.paypal.com/login. The email used for this paypal account is the **PAYPAL_RECIEVER_EMAIL**.
    a. The needed functionality is only available for business accounts and not for personal accounts.
    b. If you want to use a test business account (as you may not have a real business or you want to do some testing), go here: https://developer.paypal.com/. Click on "Log into Dashboard" in the top right corner. After this, click on sign up and fill out the form to make your developer account. This will automatically give you a test personal account and a test business account
        i. If you already have a testing business account, you can use that instead of making a new one
    c. Once logged in to your developer account, use the left side bar, scroll down to the sandbox section, and click on Accounts. On the main page, scroll down to where you see the sandbox accounts.



d. Next to the BUSINESS account, in the “Manage Accounts” column, click on the … button

   i. You should now get to see the email and a system generated password for you test business account



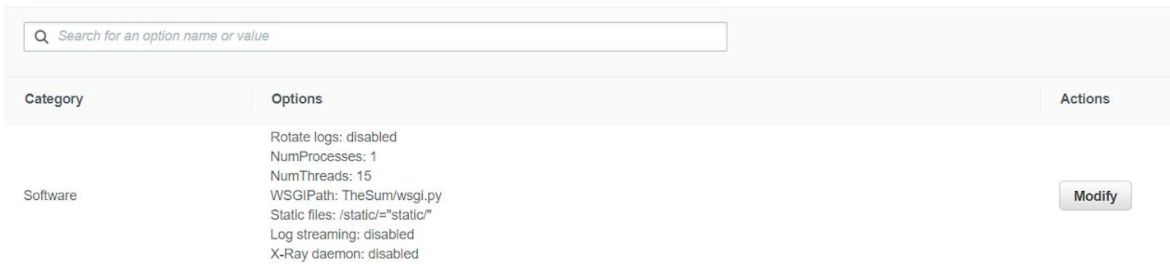
e. Use https://www.sandbox.paypal.com/us/signin to log in to your test business account (credentials found above)

24. Use the gear in the upper right corner to navigate to “Account Settings”. Under “Products and Services” select “Website payments.” Next to website preferences press “Update.” Turn Auto Return ON, and add the link to the **APP_URL** as the Return URL.
    a. For example, in step 17a, you found your **APP_URL** to be abc.com, use abc.com for the Return URL in paypal
       i. If you already have a URL set for the return URL, you do not need to change it

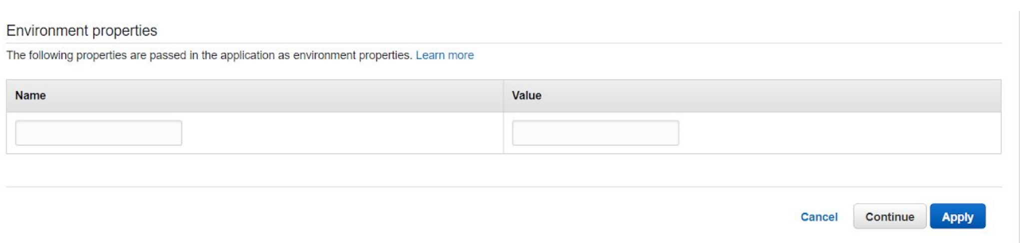
25. Turn Payment Data Transfer ON (Note that Auto Return must be turned on before Payment Data Transfer). You will need the Identity Token for the **PAYPAL_IDENTITY_TOKEN**.
    a. If you already have Payment Data Transfer set to on and your identity token is not present, set it to off and then turn it back on. Your identity token will then appear.

Setting Environment Variables

26. Now we can go back to our old tab from the start of step 18 (you may follow steps 15 through 17 again if you lost this tab)
27. Scroll up in the configuration page (which you should already be on) until you see Software. Click on the modify button. Image:



28. Scroll down until you see Environment Properties (image below)

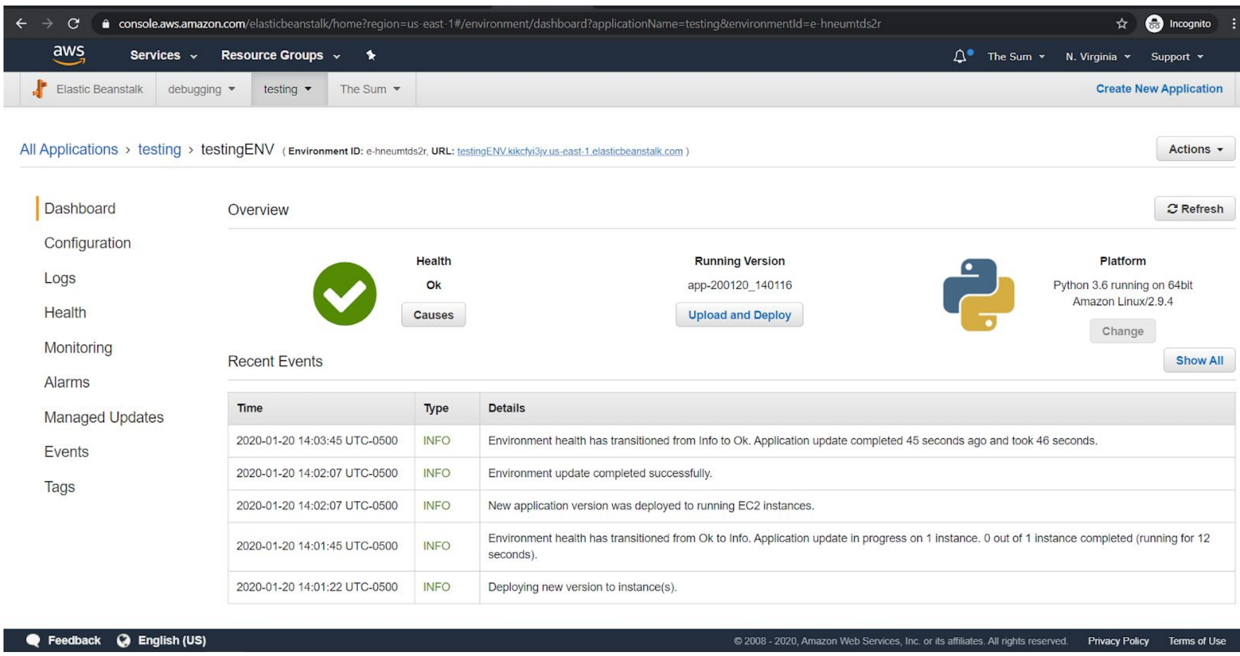


29. In environment properties, one by one, add the Name Value pairs from the table you copied at the beginning.
30. For the SEND_PDF_EMAIL environment property, if you want results emails to be sent to users, set this to anything other than OFF. If you DO NOT want results emails to be sent to users, set this to **OFF** (it must exactly match those three characters, capital, no spaces).
31. Click Apply at the bottom of this page. Finally, wait for some time and you should finally see that green checkmark!

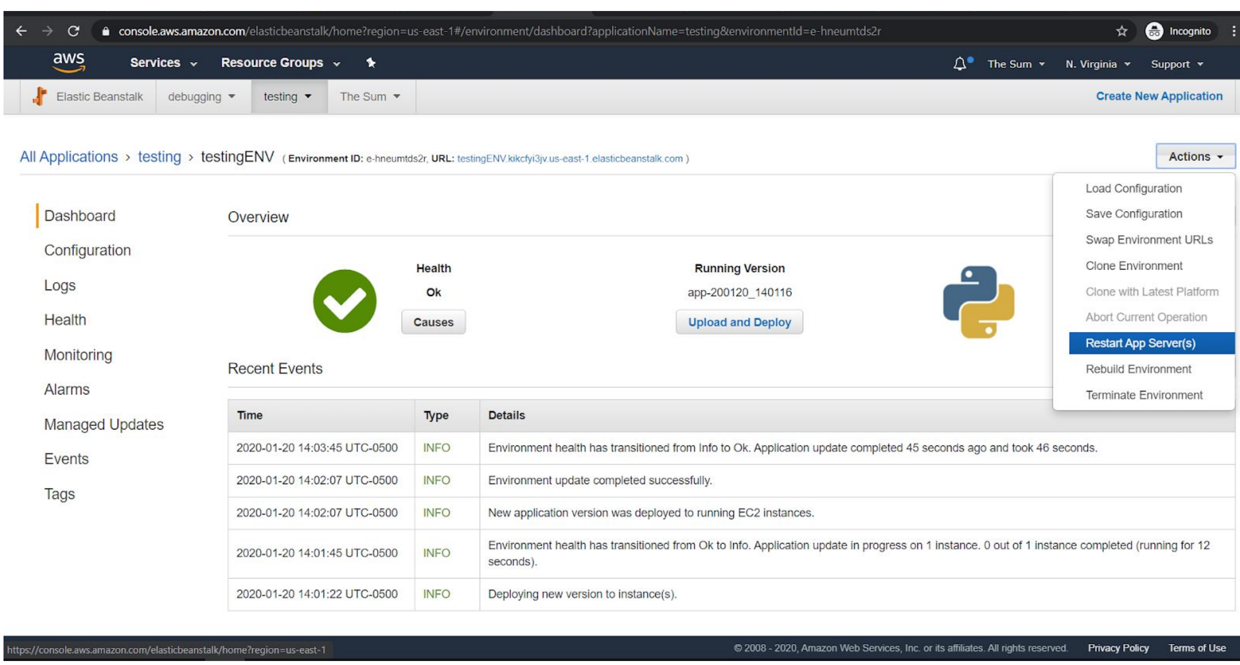
Fixing Deployment Issues

32. IMPORTANT: In step 31, if you had any issues (such as WSGI path does not exist, please do the following:

a. Navigate back to the Dashboard page for your environment. If you do not remember how to get here, please follow steps 15 through 17. It should look similar to the following:



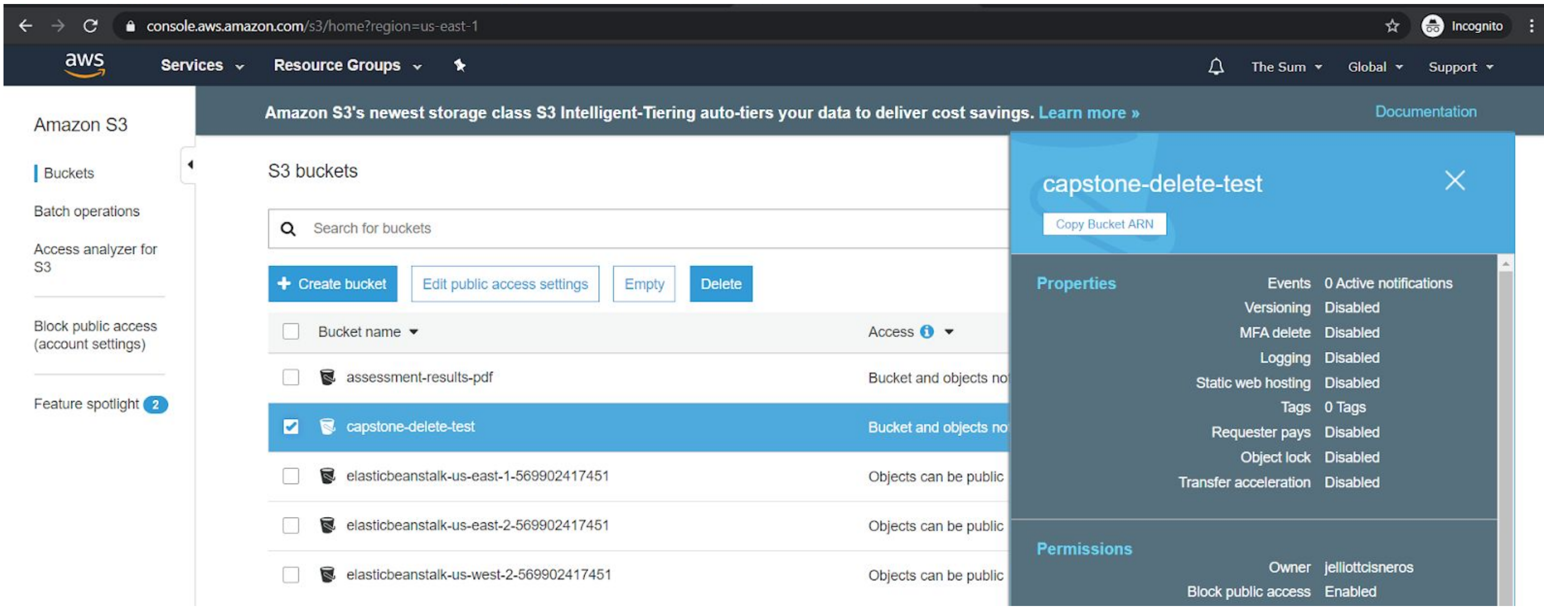
b. Under Actions, click on restart app server(s)



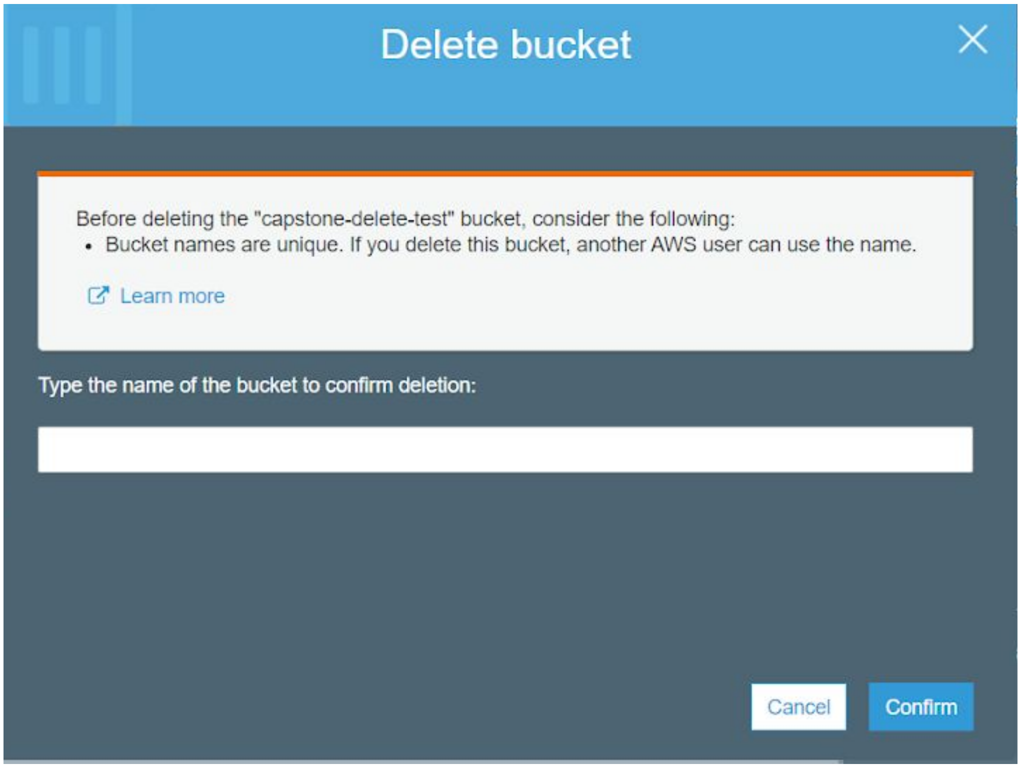
c. Click the red box that says Restart APP Server(s) to confirm this.
d. This should the WSGI path not found error (AWS has some bugs in it that we must work around). However, now going to your app you will get a 404 error or another web error.
e. Go back to your terminal that opened in the folder with the code. Type "`eb deploy`" and press enter. Once this has finished and your app has deployed then you should be done! It may take a few moments to start up, but the URL you got earlier should be accessible and you should find the app running there!

Removing the AWS Resources

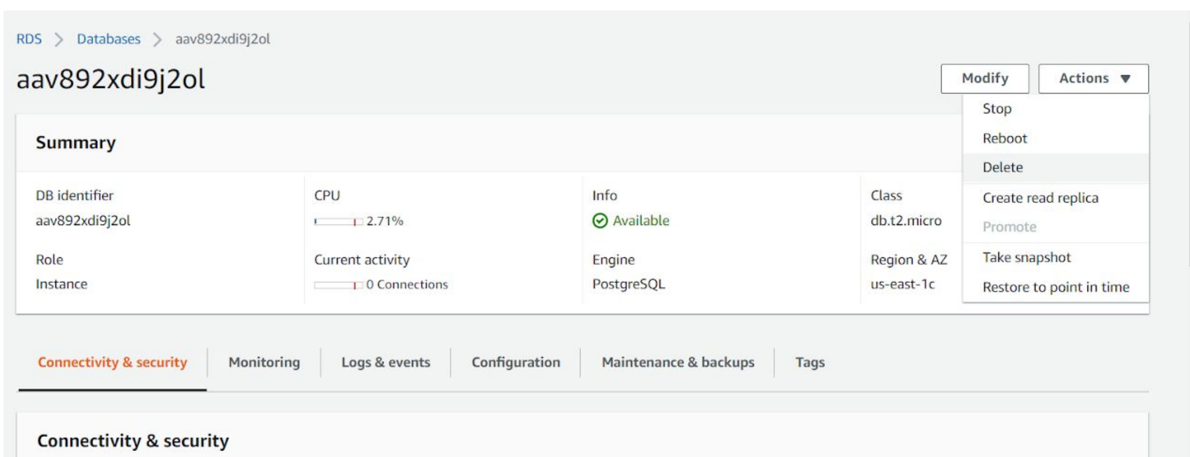
33. Deleting the S3 bucket
    a. From the main page of the AWS console, go to the main S3 bucket page like in steps 19 through 21
    b. For the bucket you want to delete, click the check box next to it and then press the delete button at the top of the list



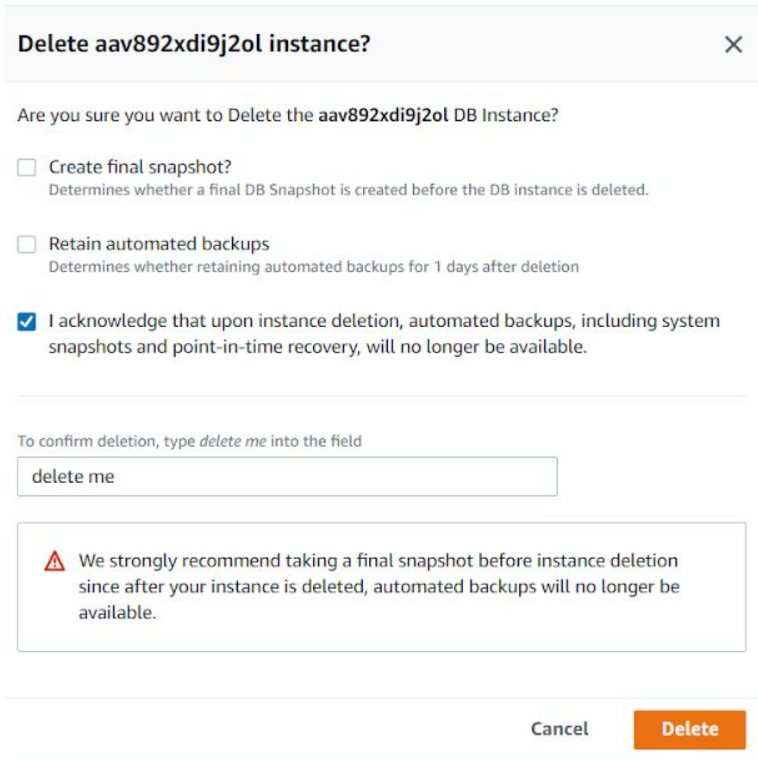
    c. You will see a screen that asks you to type the name of the bucket and then press the confirm button. Note: your environment name will appear on the confirmation page. Follow these instructions.



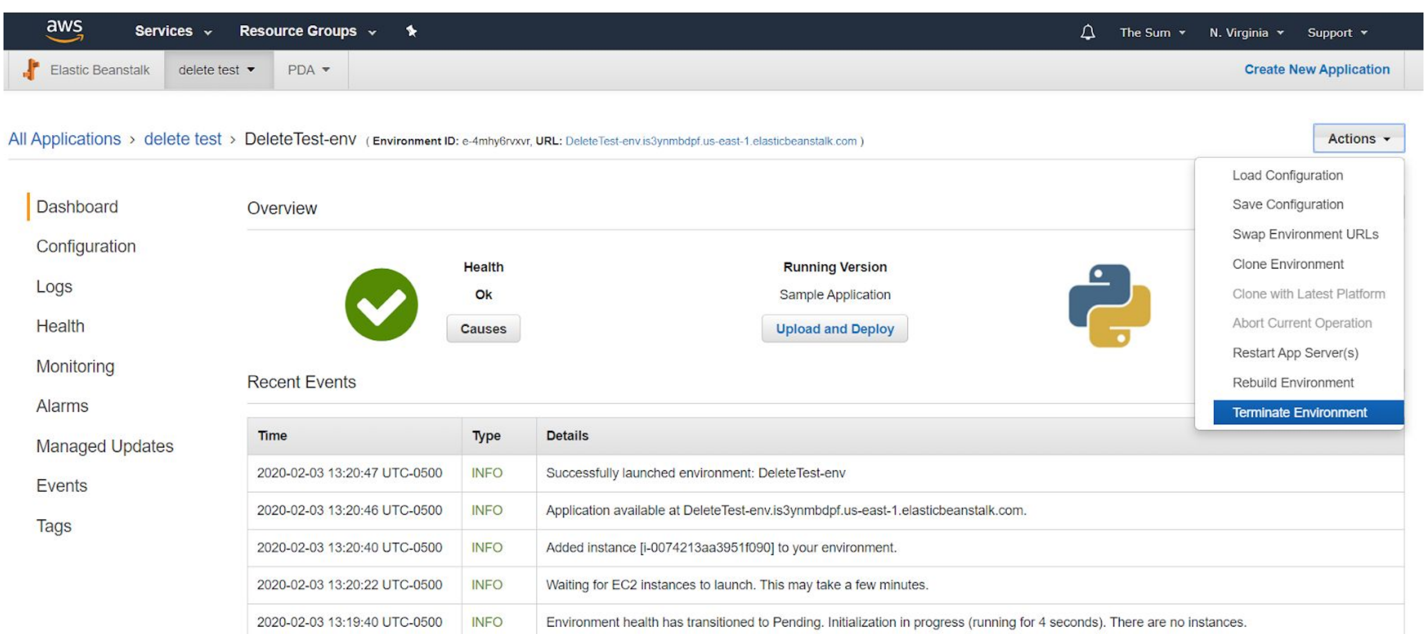
34. Deleting the database
    a. Navigate to your database like you did in step 18. If you need to get back to your environment first, follow steps 15 through 17
    b. Under the actions tab, click delete



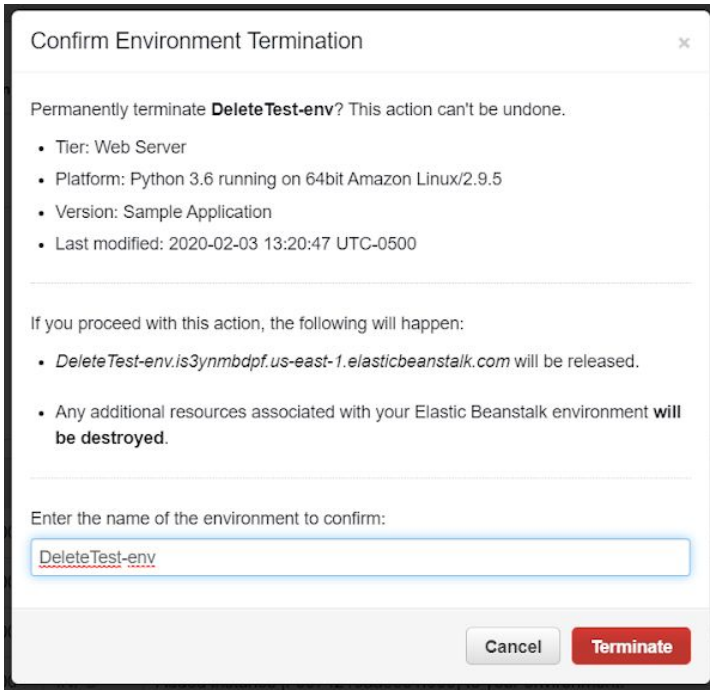
    c. You will see a new page about snapshots and asking you to confirm the deletion
    d. Uncheck “Create Final Snapshot”
    e. Check “I acknowledge that…”
    f. Type “delete me” into the field at the bottom
    g. Click Delete



35. Deleting the environment (make sure you deleted the database first)
    a. Navigate to the environments dashboard like in steps 15 through 17
    b. Click on the actions button and then click on “Terminate Environment”

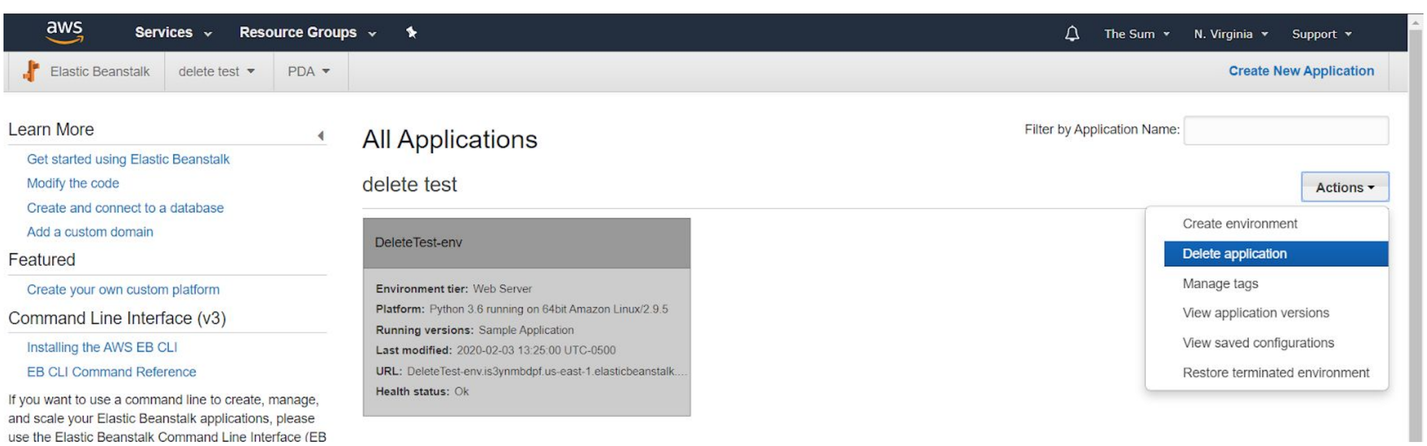


c. You will see a confirmation page. To finalize the deletion, type the environment name into the box and press the terminate button. Note: your environment name will appear on the confirmation page.

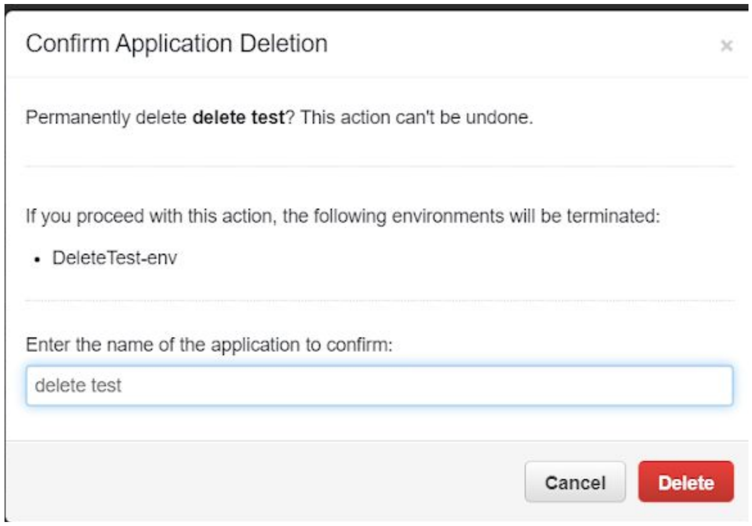


d. If you just deleted the database for the environment, it may take a few minutes before the environment gets deleted and disappears from your view

36. To delete the application (make sure you deleted the environment first
    a. Go to the applications page. If you need to, follow steps 15 and 16 to get here.
    b. For the application you want to delete, click on actions and then click on “Delete Applications”



c. You will see a confirmation page. To finalize the deletion, type the application name into the box and press the delete button. Note: your application name will appear on the confirmation page.



d. If you just deleted the environment within the application, it may take a few minutes before the applications gets deleted and disappears from your view

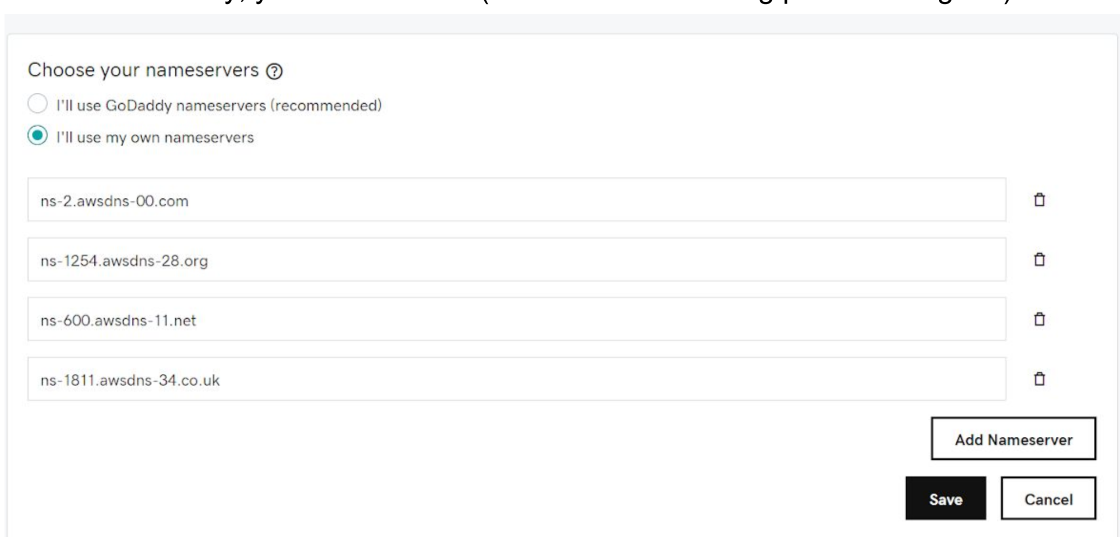
DNS and HTTPS (Optional, but must be done together)

In order to setup HTTPS, we will need to move to a new environment. At the end of this process, you will have to delete your old environment. Make sure to take a snapshot of the database so you do not lose any data!

37. We are assuming you have purchased a domain name from Go Daddy
38. First we will make a hosted zone in AWS
39. From the AWS console main page, search for and go to Route 53
40. In the dashboard on the left, click on Hosted Zones
41. Click on Create Hosted Zone and name, enter the domain name you will be using, a comment (if you want), leave type as public hosted zone, and click Create
42. Go into the hosted zone. You should see an NS record with values. We will need these values in **step 53**
43. Go to https://dcc.godaddy.com/domains/ (log in to godaddy.com first)
44. Click on the domain name you wish to use
45. Scroll down to the bottom of the page and click on "Manage DNS"
46. Under Advanced Features, you want to click on "Export Zone FIle (Unix)"
    a. In this file, you should see something like the following:
    ; A Records
    @ 600 IN A Parked

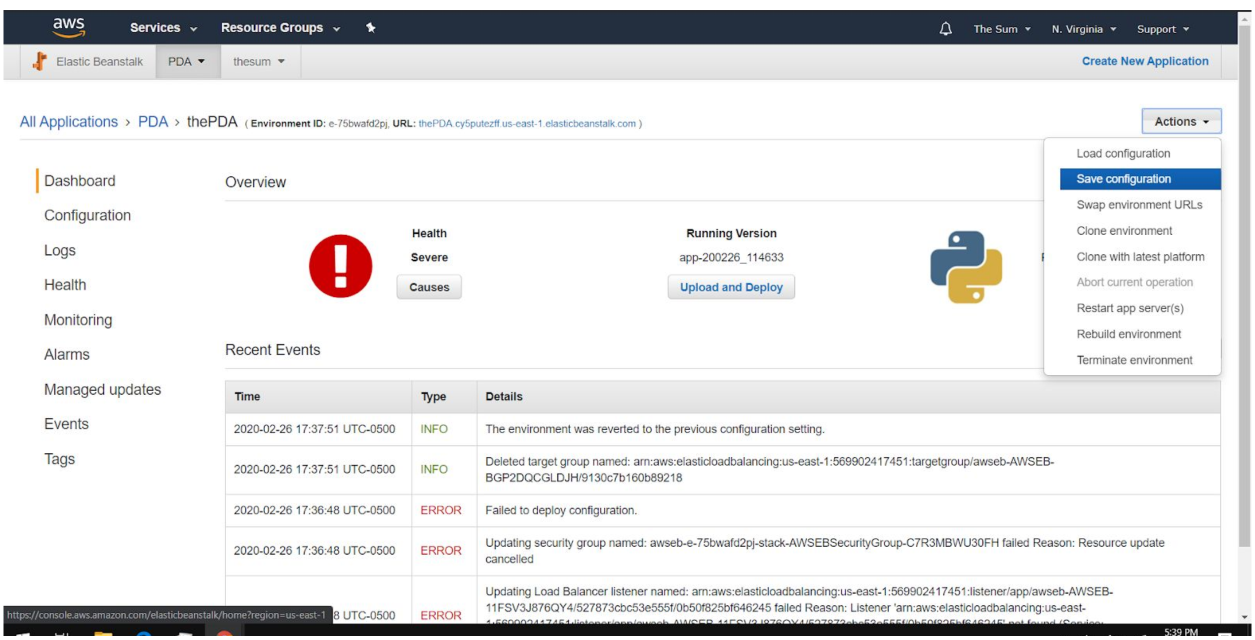
b. Change “Parked” to 1.1.1.1
c. Save the file!

47. In your AWS console, go back to where you saw the records within your hosted zone. Click on “Import Zone File” and upload our file from the previous step
48. Refresh the page and you should see new records. Click on the A record.
49. Change Alias to YES and in the alias target box, enter the URL for the app from step 17a. Click on the save button to make this change!
50. Go back to the Go Daddy page to Manage DNS
51. Scroll down to the nameservers section, click on Change
52. Click on I’ll use my own name servers. This may be towards the bottom as setting nameservers manually (advanced option)
53. Enter the values you got from **step 42**
    a. Each value from step 42 should be on its own line and will be its own nameserver here
    b. You will have to remove the trailing periods from AWS for these to work with Go Daddy
    c. Example: If you saw this in AWS:
        ns-1254.awsdns-28.org.
        ns-2.awsdns-00.com.
        ns-600.awsdns-11.net.
        ns-1811.awsdns-34.co.uk.
    Then in Go Daddy, you should enter (notice how the ending periods are gone):



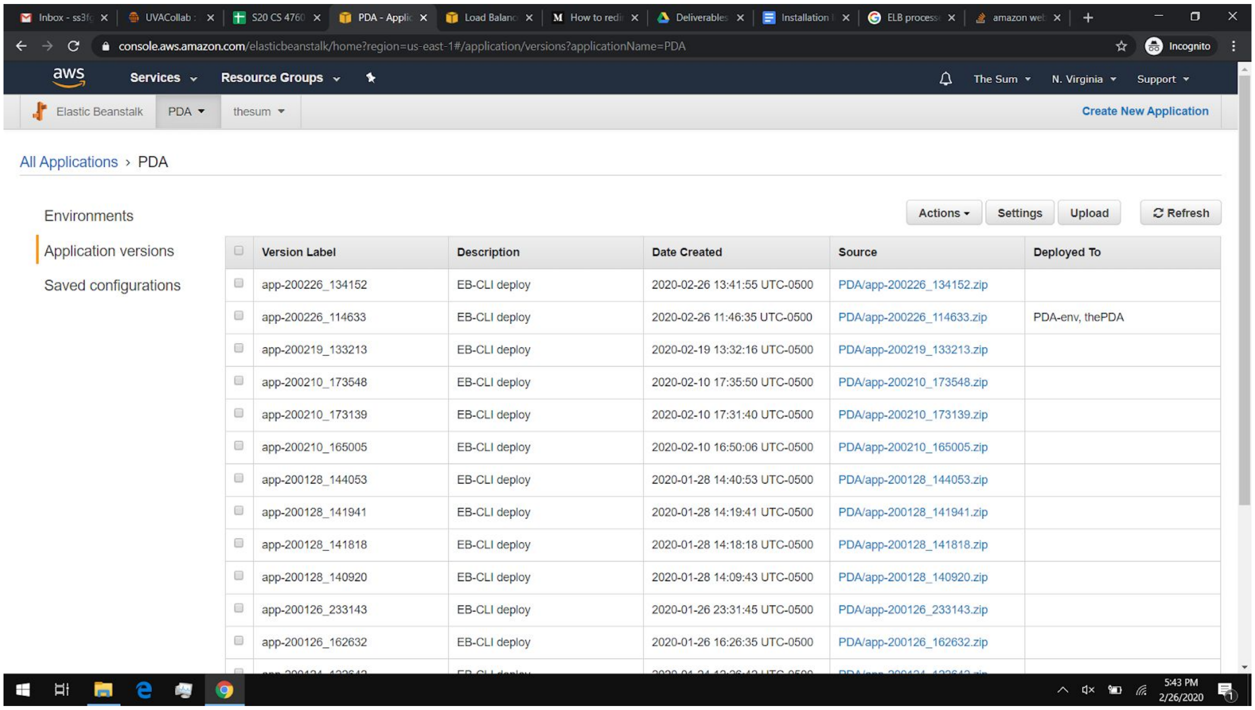
    Note: you may need to press Add Nameserver to for each nameserver you want to add over 2 nameserver. You will likely need a total of 4 nameservers!
    d. Press Save
54. Finally, go to the AWS console, go to your environment, click on configuration, and then modify for software (this will take you to where you were in step 26
55. Change the APP_URL to your domain name and save this configuration!
    a. We now have our domain name setup! All we have left to do is setup HTTPS!

56. First, we need to get a certificate for SSL/TLS
57. Once logged into the AWS console, go to https://console.aws.amazon.com/acm/home?region=us-east-1#/
58. If you do not already have a certificate for your domain name (which you probably don't right now), click on Request a certificate
59. Request a public certificate
60. When adding domain names, add <your domain name> and also add www.<you domain name>
61. Click next
62. Choose DNS validation
63. Click Next
64. You can skip adding tags, click on Review
65. Click Confirm and Request
66. Go back to https://console.aws.amazon.com/acm/home?region=us-east-1#/
67. Click on the drop down for <your domain name>
68. In the smaller box that says domain, click on the dropdowns for each domain here and click on Create Record in Route 53
69. Your certificate should be verified soon! After it's verified, follow the rest of the steps.
70. Go to your environment that you wish to move over to HTTPS. Press Actions and select "Save Configuration"
    a. You will give this configuration a name. Remember it! We will call this <saved-configuration>



71. Go up to the application level and select "Application Versions" from the bar on the left

a. Note the <version-label> for the item that is currently deployed to the environment we are recreating



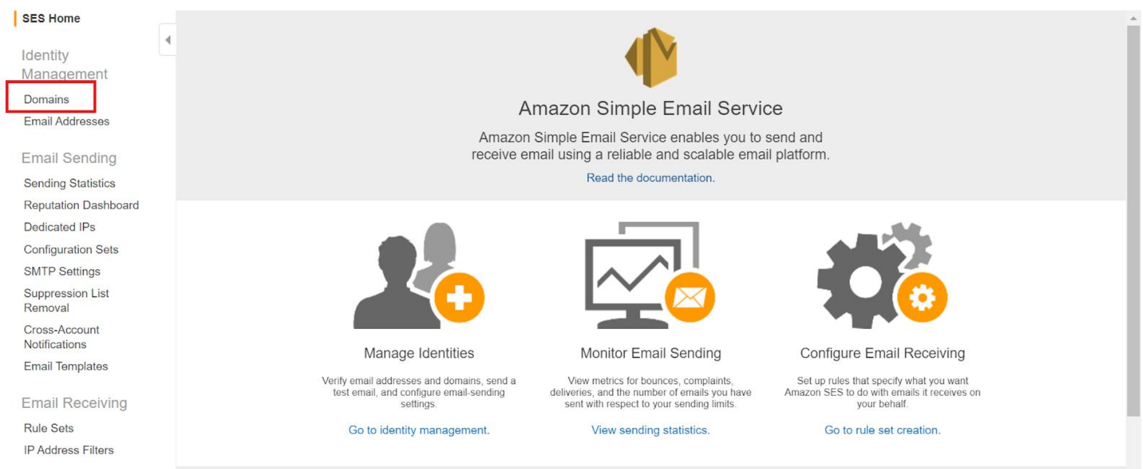
72. Go into your terminal. Find the place where you ran eb init. If you need help getting here, follow steps from earlier in the guide. You may have to run eb init again.
73. Run “eb create <new-environment-name> --elb-type application --cfg <saved-configuration> --version <version-label>”
    a. <new-environment-name> will be the name for your new environment
    b. This will run for a while, but it will give you an application load balancer which allow for us to use HTTPS
74. Once the new environment is setup, you will have to go and setup the database environment variables again (by changing the environment, we made a new database). Follow step 18 to get the database info for the new environment. Modify the RDS_HOST and RDS_DB_NAME environment variables accordingly in the software configurations for your elastic beanstalk environment. If you forgot how to get to the software configurations, see steps 15 through 17 to get to your environment in a web browser and see steps 26 through 31 to change the environment variables.
75. Follow this guide: https://medium.com/@j_cunanan05/how-to-redirect-http-to-https-in-amazon-web-services-aws-elastic-beanstalk-67f309734e81
    a. To check which load balancer to edit in the EC2 configurations, you can click on one and then go to the tags tab. It should say the environment name is the name of your new environment
76. Follow the part of the guide above to delete your old resources for your old environment!

    a. Make sure to make a snapshot of the database so you don't lose any data (if you do not have any data you want to keep, feel free to forgo the snapshot)

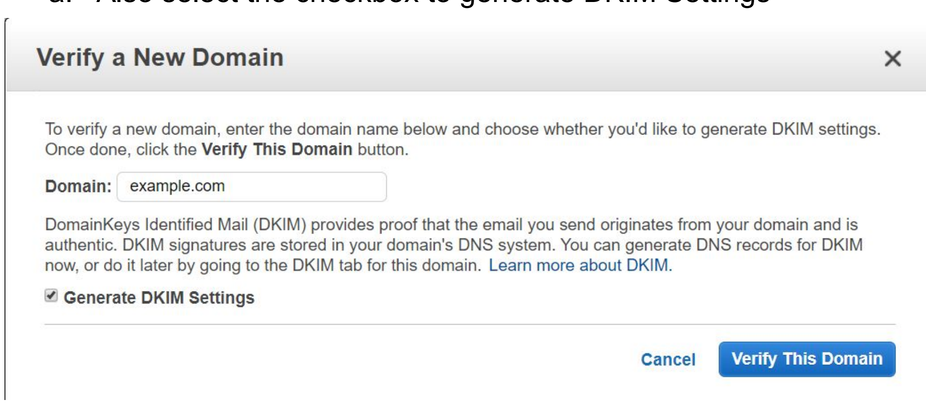
77. You may receive the WSGI issues again. If this is the case, restart the app servers like you have done previously. To redeploy, click on Upload and Deploy, then click on Application version page
78. Click the check box for the current app version (this should be the app version we used earlier in our eb create command), click actions, and click on deploy, click on the environment name you want to deploy to, and then finally click deploy
79. We now have HTTPS!

Configuring the email server using AWS Simple Email Services - for production

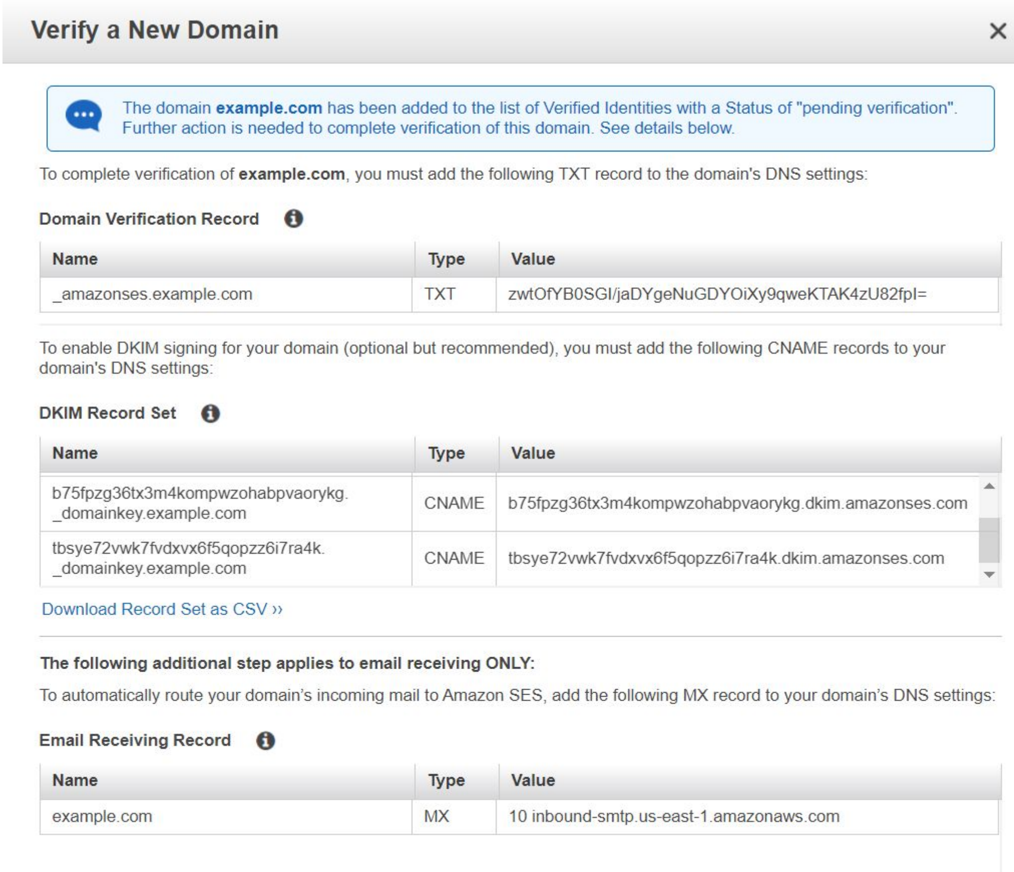
80. We are assuming that Route 53 has been set up.
81. Navigate to Amazon Simple Email Services (SES)
82. Click on "Domains" under Identity Management on the left side of the screen



83. Select "Verify a New Domain" at the top left of the center screen
84. In the "Domain" box, enter the domain name. For us, it was "thepowerofdifference.org"
    a. Also select the checkbox to generate DKIM Settings

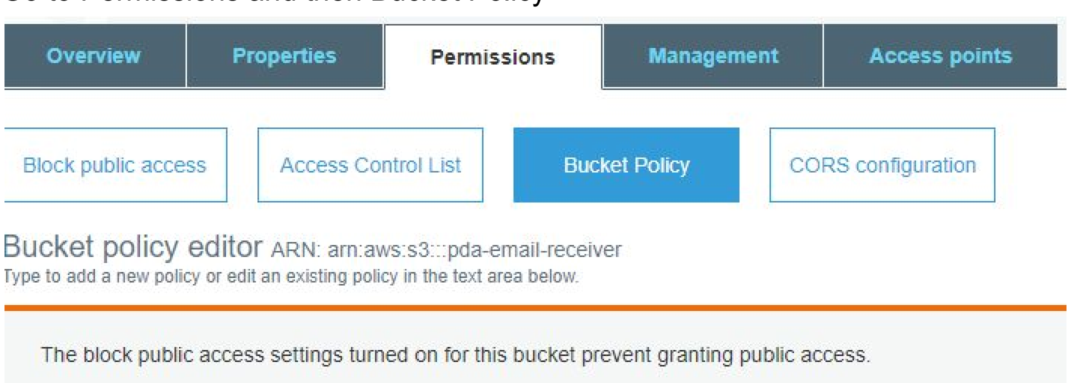


85. The result should look like this after selecting “Verifying This Domain”



86. These record sets must be updated in the domain’s DNS settings.
    a. If a button to add them automatically using Amazon Route 53 is present, select that
    b. Otherwise, navigate to Amazon Route 53, select “Hosted zones” on the left navigation pane, select the domain name that was picked in the previous section, select “Go to Record Sets,” and create record sets for each of the records in the screenshot above.
        i. Note: The records in the screenshot above will have different names and values. Use the ones that appear on your AWS page.
87. Next, set up an S3 bucket to receive emails for email verification.
    a. Navigate to Amazon S3
    b. Click on Buckets and Create a bucket
    c. On the create bucket screen, follow the instructions to create a unique name and set the region to the same region as the rest of your stuff. After clicking on the block all public access checkbox hit create bucket to create your bucket.
    d. Navigate back to Amazon Route 53 to update the record sets once again to add the following record:

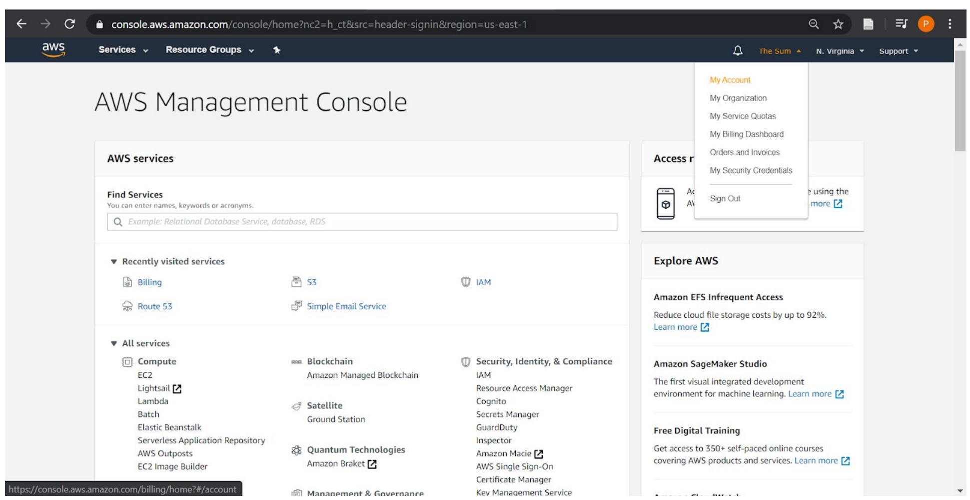
e. Notes:
   i. Leave “Name” blank. The part after name will match the domain you used when setting this up.
   ii. If you wish to use a different region, change the value to use a different region
f. After finishing up setting the records. Go back to Amazon S3 and click on the bucket you just made
g. Go to Permissions and then Bucket Policy



h. Replace bucket name and aws account id and put this in the bucket policy editor.

```
{
   "Version": "2012-10-17",
   "Statement": [
      {
         "Sid": "AllowSESPuts",
         "Effect": "Allow",
         "Principal": {
            "Service": "ses.amazonaws.com"
         },
         "Action": "s3:PutObject",
         "Resource": "arn:aws:s3:::BUCKET-NAME/*",
         "Condition": {
            "StringEquals": {
               "aws:Referer": "AWSACCOUNTID"
            }
         }
      }
   ]
}
```

i. AWS Account ID can be found by clicking on "My Account" and looking under Account Settings at the top



ii. Replace BUCKET-NAME with the name provided for the S3 bucket.

i. Hit the save button to save the policy.

88. Email Verification
   a. Go back to Amazon SES and click on "Email Addresses" which is under "Domains" in the Identity Management section
   b. Select verify a new email address and enter the email address you want to be verified.
      i. Note: Use the domain name you verified in the last step (e.g. info@example.com)
   c. A verification email will be sent to this domain.

d. Navigate into the S3 bucket by going to AWS S3 and selecting the S3 bucket that was created in the previous step.
    i. There should be two emails. One is AWS_SES_SETUP_NOTIFICATION and can be ignored. The other will appear to be random numbers and letters and should have a very recent last modified time.

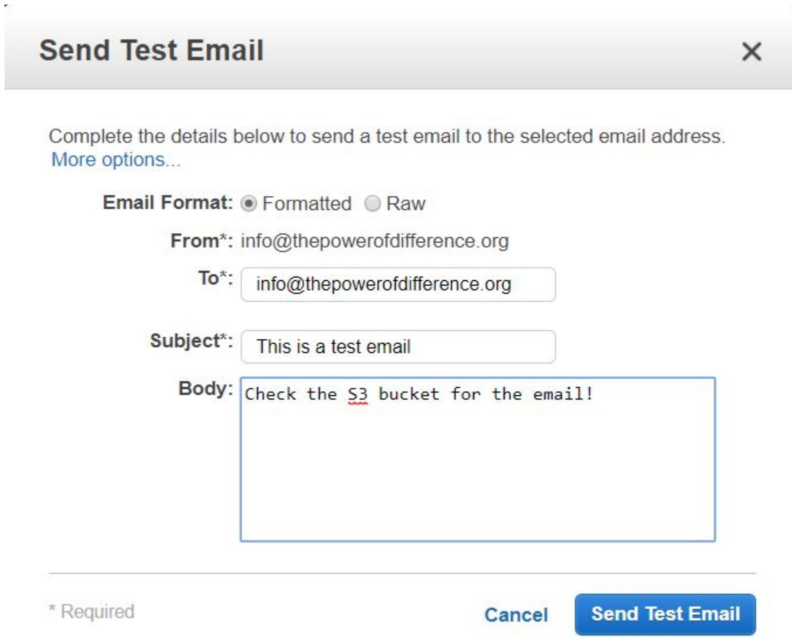
Name	Last modified	Size	Storage class
0667uilggos3kbdmk32baputi5o3o94laaq2c601	Mar 1, 2020 2:02:56 PM GMT-0500	5.2 KB	Standard

e. Select this email and click “Download.” Open the file that has been downloaded using a text editor, such as Notepad. You may have to find the downloaded file in the downloads folder before right clicking the file and selecting “Open with” to view the contents
f. Scroll down to the bottom of the file until you find the email contents, which should look something like this

```
From: Amazon Web Services <no-reply-aws@amazon.com>
To: info@thepowerofdifference.org
Message-ID: <01000170974d31cc-e5637465-fed8-4301-9993-607ac230db32-000000@email.amazonses.com>
Subject: =?UTF-8?Q?Amazon_Web_Services_=E2=80=93_Email_Address_Verifica?=
 =?UTF-8?Q?tion_Request_in_region_US_East_(N._Virginia)?=
MIME-Version: 1.0
Content-Type: text/plain; charset=utf-8
Content-Transfer-Encoding: 7bit
X-SES-Outgoing: 2020.03.01-54.240.13.38
Feedback-ID: 1.us-east-1.ZHcGJK6s+x+i9lRHKog4RW3tECwWIf1xzTYCZyUaiec=:AmazonSES

Dear Amazon Web Services Customer,

We have received a request to authorize this email address for use with Amazon SES and Amazon Pinpoint in region US East (N. Virginia). If you requested
this verification, please go to the following URL to confirm that you are authorized to use this email address:

https://email-verification.us-east-1.amazonaws.com/?Context=569902417451&X-Amz-Date=20200301T181303Z&Identity.IdentityName=info
%40thepowerofdifference.org&X-Amz-Algorithm=AWS4-HMAC-SHA256&Identity.IdentityType=EmailAddress&X-Amz-SignedHeaders=host&X-Amz-
Credential=AKIAJR7UYJEP5GNMLX6A%2F20200301%2Fus-east-1%2Fses%2Faws4_request&Operation=ConfirmVerification&Namespace=Bacon&X-Amz-
Signature=fb99680239c231d820a790c101072749fa6b8e3b9c00e88371c595a573f521d0

Your request will not be processed unless you confirm the address using this URL. This link expires 24 hours after your original verification request.

If you did NOT request to verify this email address, do not click on the link. Please note that many times, the situation isn't a phishing attempt, but
either a misunderstanding of how to use our service, or someone setting up email-sending capabilities on your behalf as part of a legitimate service, but
without having fully communicated the procedure first. If you are still concerned, please forward this notification to aws-email-domain-
verification@amazon.com and let us know in the forward that you did not request the verification.

To learn more about sending email from Amazon Web Services, please refer to the Amazon SES Developer Guide at
http://docs.aws.amazon.com/ses/latest/DeveloperGuide/Welcome.html and Amazon Pinpoint Developer Guide at
http://docs.aws.amazon.com/pinpoint/latest/userguide/welcome.html.

Sincerely,

The Amazon Web Services Team.
```

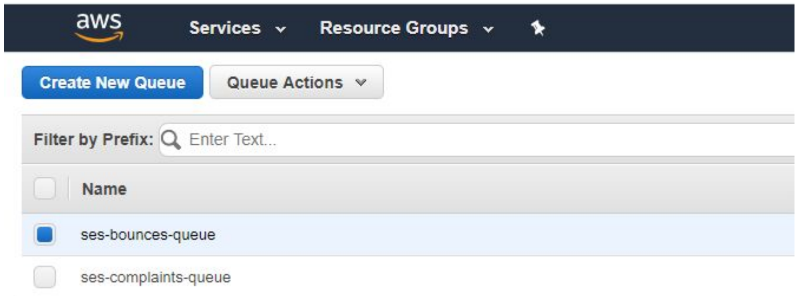
g. Copy paste the email verification link into a browser to successfully verify the email account.

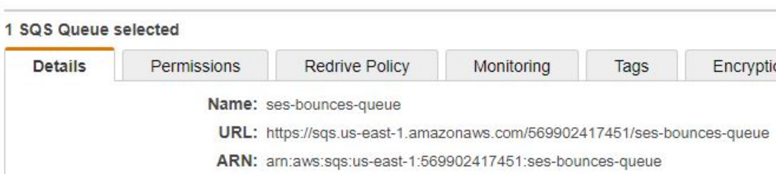
89. Optional - Send a test message to your domain
    a. Navigate to Amazon SES, select email addresses, choose the email address that was verified in the previous step, and click on “Send a Test Email”
    b. In the “To*:” section of the email, make sure to use your verified domain.



c. Navigate back to Amazon S3 and view the S3 bucket for receiving emails.
d. A new email should appear with the contents above.

90. To send emails to unverified emails, a support request must be submitted to get the application out of sandbox mode.
    a. Navigate to the AWS Support Center and select on “Create Case” under Open support cases
    b. In the Create Case Info, select “Service limit increase”
    c. Under Limit type, select “SES Sending Limits”
    d. In the contents of the support request, describe the system, what emails are needed for, and how you will comply with AWS SES standards.
    e. It may take a while for the support request to be handled. Once handled, the application will be out of sandbox mode and you can send emails to any email address.
91. Handling Bounces and Complaints:
    a. Go to Amazon Simple Queue Service (SQS) click create a new queue.
    b. Name the new queue something like ses-bounces-queue, set it to “standard queue,” and then create the queue.
    c. Create another queue with the same parameters called ses-complaints-queue.
    d. Look at each of the queues and write down their arn values, which can be found at the bottom of the page.





e. Next head to Amazon Simple Notification Service (SNS). Select topics on the left and create a new topic.
f. Use “ses-bounces-topic” as the name. Leave all other settings as their defaults, then create the topic.
g. Create another topic called “ses-complaints-topic”
h. Click on the ses-bounces-topic and click “Create Subscription”
i. In this next page set protocol to SQS. Then set the end point the ARN value for ses-bounces-queue from earlier. Then create the subscription.
j. Do steps h and i again for ses-complaints-topic, and set the end point to the ARN value for ses-complaints-queue from earlier.
k. After this, go back to Amazon SES and click on domains.
l. Select your domain, and hit “View Details”
m. Go to the notification tab and hit edit notifications.
n. Set bounces to ses-bounces-topic and complaints to ses-complaints-topic and hit save config.

92. Change environment variable to use AWS email server (production)
    a. Go to the environment variable on AWS. Refer to Step 26 to remember how to reach this page and change env to production if it’s not already.